

The Thing Nobody Chose

*An honest moral framework for people who don't believe in cosmic
authority*

Brennan Kelley

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Prologue — The Thing Nobody Chose

September 2022. Rachel Kudrna is a fourth-year at Washington State University's Elson S. Floyd College of Medicine, and it's the month the Electronic Residency Application Service opens for the year's applicants. The portal looks like a tax form: checkboxes next to program names, a running total in the corner, a submit button that doesn't move until the total is paid.

The average U.S. medical graduate that cycle will submit around seventy of those checkboxes. A decade earlier, the average was roughly half that. Seventy applications, at the fee schedule in effect that cycle, comes to about sixteen hundred dollars before a single interview gets booked, and interview travel is separate. Sixteen hundred dollars is a pretty big chunk of change for a fourth-year who has already taken out loans to get this far. For the more competitive specialties, interview travel alone can run past ten thousand dollars. Flights, hotel rooms, a suit that has to look the same in Cleveland as it did in Houston. The fourteenth Tuesday in a row spent explaining, in a different building, why this particular hospital is the one she has always wanted to work at.

The programs feel it from the other side of the same screen. One program, reviewing the flood of applications that arrives every autumn, put the cost in lost faculty clinical time at close to

thirty-seven hundred dollars per applicant interviewed. That is one program's number, not a national one — but the clinic hours are real, spent reading files instead of seeing patients. The typical number of interviews a successful applicant actually completes has crept up too, from ten a decade ago to thirteen now. The people on the other end of the portal are reading more files to fill the same number of chairs.

Kudrna puts it simply: "There's tons of anxiety. It feels like after working so hard to become a physician, it could all still fall apart."

Down the hall, or somewhere equivalent to a hall in a college of medicine that spans a state, is Leila Harrison, WSU's senior associate dean for admissions and student affairs. Part of her job is telling students not to do what Kudrna and her classmates are, in September, doing anyway. Harrison advises against padding the list. A program can only train so many residents; seventy applications from one student don't create a seventy-first slot to fill. She gives the advice every year.

Every year the number goes up anyway.

Here is how Harrison herself describes watching it happen: "It's heartbreaking because this involves so much money and effort. Even though we advise students not to do it, they do it anyway."

Nobody in this is doing anything unreasonable.

Kudrna is applying to enough programs to protect herself against the chance that thirteen of them say no. Harrison is giving sound advice to a room full of people who cannot individually afford to take it, because the protection only works if you have more of it than the person next to you. The programs, on their end, are drowning in files, because every other program is also drowning in files, for the same reason, at the same time, every

September. It is the last week of the month. Kudrna has the portal open on her laptop, the cursor sitting in the box for program number forty, then forty-one. Somewhere the total is still climbing.

If you're like most people reading this, part of you is already doing what I did when I first heard about it: scanning for the one person in the story who's actually being unreasonable. The greedy program. The neurotic student. The dean who should just tell them no already. Hold that instinct a second.

I didn't go looking for this because of my niece, actually. I went down a different rabbit hole first, reading about residency admissions for an unrelated reason, and it was there I found Harrison's line. I went back afterward and found Kudrna's cycle, and only then realized my own family was living a version of the same thing three states away.

I have a niece in almost exactly this position this year, not Kudrna, a different fourth-year, in a different state, doing the same arithmetic on a different laptop. Sarah is the loud one at Thanksgiving, the one who has never once let a relative forget she beat them at cards. She's applying to something like seventy programs herself.

Seventy.

Not because she wants to work at seventy hospitals; she'd be happy at six, thrilled at three. But everyone applies to seventy now, and if she applies to fewer than everyone else, she's the one standing there in March with nothing.

It's late while I write this part, coffee gone cold next to the keyboard, the kind of hour where a person's guard comes down.

I want to be honest about something here, because it's the kind of thing a person is tempted to leave out. My first reaction, when Sarah explained the seventy-programs arithmetic to me over Christmas, was not sympathy. It was closer to irritation. I

remember thinking, privately, something like this: this generation and their neuroses, just pick a dozen good ones and relax. I knew it was a stupid thing to think within about four seconds of thinking it. I'm putting it on the record anyway, because it's the instinct almost everybody has when they meet a story like this one — including people who should know better, including me. The instinct is: somebody here is being unreasonable. Find them. Blame them. So I spent a week actually trying to find that somebody. Sarah wasn't it. Neither was Harrison. I couldn't even make it stick to the programs. I came up almost empty. The one thing I found sits upstream of all three, in an act of Congress rather than in any of their offices, and it doesn't let anyone in this story off the hook.

Every person I could locate in the chain was doing the sane, defensible, individually correct thing. The programs aren't being greedy; an empty seat in July is a real cost, and they're not wrong to guard against it. The applicants aren't being neurotic; the student who applies to twenty-five programs like it's 2011 might genuinely not match anywhere, and for a fourth-year, not matching isn't a minor setback. It's close to a small career death. The deans aren't lying when they tell a room full of anxious twenty-six-year-olds not to do this; the advice is true, for any one student, taken alone. Everyone is right. And several thousand people being right, all at once, in the same direction, is how you end up with a system where a woman who can't really afford it pays sixteen hundred dollars to apply to hospitals she will never see the inside of, on the chance that one of them says yes. It's a pattern I recognize in myself now, generally: impatience first, the slower work of actually checking a distant second. I'm not proud of the order.

This had gone on, unaddressed, for about fifteen years, deans giving the same advice every September and watching the number climb anyway, before anyone tried anything at all.

They did try to fix part of it, eventually. The application service belongs to AAMC, which collects a fee on every checkbox a student ticks, and it added a way for students to flag a handful of programs as their real top choices. Now a program could tell the difference between someone who meant it and someone blanket-applying to everything with a pulse. Applications did come down, in some specialties by close to half; by one tally, dermatology alone went from around seventy applications per applicant to about forty in two years. I think that's real progress, for what it's worth. Whether it holds is another question.

Because the same shape simply moved one level down. Now it's research. By one accounting, neurosurgery applicants report nearly thirty research activities each, on average, and in a recent national survey, research sits among the factors program directors most often use to decide who gets an interview. In otolaryngology, the research items listed by successful applicants roughly doubled in six years, from about eight to about seventeen. The bar didn't just rise. It rose out from under the people trying to clear it. Students are stacking their applications with research they don't especially care about, instead of, say, getting better at talking to actual patients, because research is the new thing that separates you from the pile. A medical student named Anmol Shrestha wrote, under his own name, about skipping a class in medical Spanish to keep doing research instead. A clinically useful skill, traded for a line on a form — and he was honest enough to say so in print. You plug one hole and the water finds a different wall.

I keep thinking about this because it isn't really about medicine. Ask anyone who hires software engineers what

happened to their interview process over the past decade. One coding challenge became two take-home projects, then a case-study presentation to a panel that used to just be a single manager on a phone call. Plumbers don't do this to each other. Neither, as far as I've noticed, do electricians, and I have a half-built theory about licensing boards that I'm not going to drag in here. But I don't think it's a coincidence that these stories keep turning up in the same shape, in fields that have nothing else in common. Every manager involved thought the whole thing had turned into a mess, and every manager kept doing it, because the manager who stopped first would be the one who hired worse.

There are textbook names for pieces of this, and not one of them tells you who to be angry at. Plenty of people do fine out of arrangements like this one, so "everyone loses" isn't right. It isn't an accident either, which rules out "nobody's fault." For about a day I tried "the market's fault," until I noticed there wasn't really a market here in any normal sense, just a lot of separate, reasonable people responding to each other.

One piece of it does have an owner, and it's the piece I found that week. The number of residency positions a hospital can fund isn't set by how many exam rooms it has; it's set by a federal cap Congress froze in the Balanced Budget Act of 1997, acting on its own advisors' prediction of a coming doctor surplus that never arrived. Medical schools kept expanding for twenty years after that. Congress's first thaw didn't come until 2021: a thousand new slots, phased in at two hundred a year. That explains the size of the pot everyone in this story is fighting over. It doesn't explain why Kudrna is still filling out box forty-one instead of stopping at twelve — that part of the trap runs entirely on people who never voted on the cap and mostly don't know it exists.

It happens the same way, on schedule, every time, like something with a mechanism. It's a weird thing to sit with, honestly. I'm confident about that part. I'm much less sure what anyone is supposed to do about it.

What I keep coming back to is smaller and harder than either of those: how do you hold anyone responsible for a result that nobody chose and everybody produced together? Who, exactly, do you get angry at?

Who do you forgive?

Part One — The Trap

1 • Everyone Hates This

I didn't have a good answer to that question when I left it sitting at the end of the last chapter. The closest thing I have to one starts thirty years earlier, on a wharf in Newfoundland, with a man in a wool coat getting yelled at by a crowd that had already lost.

Bay Bulls is a fishing community about thirty kilometres south of St. John's, and on the first of July, 1992, a crowd of roughly a thousand people filled the wharf there for what was supposed to be a Canada Day event. Fishers and plant workers, wives and husbands, kids dressed for the holiday. The air in a harbor like that smells the way it has for five hundred years of processing fish: salt and diesel and something underneath both. John Crosbie, the federal fisheries minister, came to speak. He was a Newfoundlander, from St. John's West, but he arrived in a navy wool coat and a tie, dressed for Ottawa. He stood among a crowd in ball caps and bright nylon jackets who by then knew, or badly suspected, what he was going to tell them the next day.

They didn't wait for the next day. They started yelling at him on the wharf.

Crosbie told the crowd there was no need to abuse him. A voice came back. Someone asked whether he was a Newfoundlander or whether Ottawa mattered more to him now. Crosbie answered:

“Why are you yelling at me? I didn’t take the fish from the goddamn water, so don’t go abusing me.”

A voice off to the side asked, well, who took it then. Another answered him directly: you and your goddamn people took it. You and your people took it.

The next day, July 2, Crosbie announced a moratorium on the Northern cod fishery. He called it temporary, roughly two years. It put about thirty thousand fishers and plant workers out of work in a single announcement, the largest mass layoff in Canadian history, and it emptied out more than four hundred communities that had no other major employer. The moratorium did not lift, even partially, until 2024.

Crosbie went back to St. John’s that evening, and the crowd went back to whatever was left of their Canada Day.

Five hundred years, this fishery had run before that morning. When John Cabot’s crew came back in 1497, the reports that reached Europe — a Milanese envoy in London wrote them down that December — described fish so thick that sailors claimed you could slow a ship just by sailing it through them. It took about fifteen years, at the end, to empty the water those reports had described.

Nate, that’s the whole scene, or close to it. I found it a few weeks ago. I’d gone down a research hole about fisheries for an unrelated reason and couldn’t climb back out once I got there. You’re going to say I’m doing the thing again where I read one depressing case and then need to call somebody and explain it, and you’re right, that is exactly what’s happening. You’re the one I called because you’ll actually argue back instead of saying “huh, interesting” and changing the subject to the Sox. You did that once standing in your driveway, cordless drill still running in your hand, and somehow won the argument anyway. And you’ve been

wrong often enough in these calls that I still owe you at least one real apology, which isn't coming today either. So here it comes. Sorry in advance.

Here's what I've been trying to work out: why bad things keep happening that nobody, individually, wanted. Not bad things caused by bad people. We've got a whole vocabulary for that, and mostly it's the right vocabulary. I mean the other kind. The kind where you interview everybody involved afterward, and every one of them was doing something defensible, sometimes something smart. The sum of all that defensible behavior is a disaster nobody would have signed up for if you'd shown them the ending first.

I used to think that kind of case was rare, a curiosity, the sort of thing you file next to two-headed calves and other oddities. I don't think that anymore. I think it's one of the engines behind a lot of the worst stuff that happens — I haven't weighed it yet against the uglier kind, the kind with an actual villain in it, and that's a different reckoning for a later chapter. But I think this shape does more damage than its reputation as a curiosity suggests, and the rest of this book is the case for that, not a fact I'm handing you already proven. It has a shape you can learn to recognize the way you can look at a foundation and know the deck is going to sag in four years even though nothing looks wrong yet. Once you've seen it once, you start seeing the outline of it everywhere, in situations with nothing else in common: a fishery, a bidding war, a group chat where everyone's waiting for somebody else to admit the plan fell through.

I want to tell you it was the crowd on that wharf. The greedy fishermen, emptying the ocean to fill their own boats, the villain the story hands you before you've even finished the first paragraph.

It wasn't.

I wanted it to be the fishermen. Not because I have anything against fishermen. I've never met one that I know of. It's because a story with an identifiable villain is so much easier to write, and easier to feel something about, than a story where the honest answer turns out to be a math error nobody caught for two years, compounded by an agency that didn't want to say the quiet part out loud in an election year. I'd read the crowd's anger on the wharf, read Crosbie's line back to them, and some part of me had already cast the roles before I'd read a page of the actual research. They're the mob. He's the bureaucrat catching blame that should have been theirs. It's a good story. It has a shape I already knew how to tell, from a hundred other stories with the same cast.

And it's wrong. Or at least wrong enough that a fisheries sociologist would stop reading me after the first page if I wrote it that way. So I spent most of a week doing the slower, less satisfying work of actually checking, and the case kept refusing to hand me the villain I'd shown up expecting. It's a reflex almost everyone has when a story lands on them in roughly this shape, including plenty of people who write books about ethics for a living. Including me. Find whoever's fault it looks like. Stop looking once you've found them. That's the whole move. It's fast, and it usually feels like clarity rather than laziness while it's happening. I went looking anyway, slower, for longer than I wanted to. Here's what I actually found instead.

Local inshore fishermen, the ones running small boats and fixed gear close to shore, get cleared in the peer-reviewed literature. A sociologist named Peter Sinclair tracked the collapse as closely as anyone did. He put it almost that bluntly: overfishing by local inshore fishers "must be discounted as a major factor" in the collapse — even as they bore the brunt of what followed.

They're closer to the people who show up first at a fire and get blamed for it than to anyone who lit it.

So who did it. Before 1977, mostly foreign trawlers, working water Canada didn't yet control. Then Canada declared a two-hundred-mile economic zone and pushed the foreign fleet out, and its own offshore dragger fleet expanded fast to take up the slack. That's the part almost nobody remembers: the dominant driver after 1977 wasn't some outside villain. It was Canada's own industrial fleet, each company running boats it was, from where it sat, entirely reasonable to run.

Meanwhile the government's own model for counting fish was quietly broken by something almost boring, a failure fisheries researchers would later document across more than one collapsed stock: it used how many fish the boats were catching as a stand-in for how many fish were actually still out there. That measure lies to you once the gear gets better. Sonar and net technology kept improving, so a shrinking number of fish got caught at the same steady rate for years, the gauge reading full while the level underneath it kept dropping, right up until it didn't.

In 1987 the inshore fleet paid for an independent audit out of its own pocket, because they didn't trust the government's own science advisory committee, a body called CAFSAC, to catch its own mistake. Their boats were finding fewer fish than the official numbers said should be there. The audit, run by an American fisheries scientist named D. L. Alverson, went further than a stale gauge — it found a specific mathematical fault built into the model itself, one that had been quietly expanding its own error for years. Alverson described the search for it, years later: "it took several years to find it, and by then, the stocks were on the downturn. I think it came as a shock to all of the fishermen and some of the scientists." By 1989 CAFSAC's own scientists were

recommending catch cuts of more than half. The government didn't cut that far. The quota Crosbie set for 1992, 187,969 tonnes, was higher than the entire industry's catch the year before, 129,033 tonnes. He authorized more fish to be taken than anyone had managed to find and land the previous season.

None of this was hidden, exactly, which is the part I keep circling back to. A researcher named Alan Finlayson studied the stock-assessment fights. He concluded that the inshore fleet's warnings were filtered out long before they reached anyone with the authority to act on them. When real doubt surfaced in the early eighties, and again when Alverson's report landed in 1987, the people running the department mostly ignored it or told the naysayers to settle down. The Alverson report was tabled at a federal-provincial meeting that same year, read into the record, filed. It didn't disappear. It just didn't move anything. Nobody sat in a room and decided to bury a warning. It was more like a series of people, each reasonably busy and reasonably confident in the system they worked inside, each deciding this particular red flag could wait for the next budget cycle, until there were no cycles left.

There's a name for what almost happened here. It's the most famous name in this whole territory, and also the most misused. In 1968 a biologist named Garrett Hardin published an essay in *Science* called "The Tragedy of the Commons." His example was herdsmen sharing a pasture. Each one gets the full benefit of adding another animal to his own herd and only a fraction of the cost, because the overgrazing gets shared by everyone. Run that arithmetic and the pasture gets ruined, not because any herdsman is stupid or greedy, but because the incentive facing each one, individually, points the same direction as everyone else's.

I'm going to spend a little longer on Hardin than the argument strictly needs, because the man himself is a mess in a way that

rarely makes it into the summary. He didn't even invent the herdsman example. He borrowed it from William Forster Lloyd, an Oxford economist who'd used almost the same overgrazed-pasture case in lectures back in 1833, and says so himself in the essay. I find that detail more interesting than I probably should: a hundred and thirty-five years between the first telling and the famous one, and the pasture hadn't changed at all. And the 1968 essay wasn't really about pastures, or fish. The herdsman was the warm-up act; the essay's actual center was human population growth, the argument the pasture was built to sell. I'm not going to wade into that one here — it deserves its own treatment, not a paragraph borrowed from someone else's chapter.

Cod fits Hardin's herdsman shape closely enough that "tragedy of the commons" is the label you'll see stamped on this story in nine articles out of ten. Here's the thing almost none of those nine articles mention: Hardin himself walked part of it back. Thirty years after the original essay, in a follow-up piece in the same journal, he admitted the paper had a hole in it. Not a small one, in his own accounting:

"the weightiest mistake in my synthesizing paper was the omission of the modifying adjective 'unmanaged.'"

A managed commons, he went on, can be run well or run badly, the same as a market or a state can. The devil is in the details, his phrase, not mine. It's an *unmanaged* commons, one nobody is governing, where ruin gets built into the arithmetic. Elinor Ostrom spent a career documenting exactly how communities do the managing. Swiss alpine pastures. Filipino irrigation systems. A water court in Spain I'll tell you about properly a few chapters from now. None of them collapsed the way Hardin's pasture did. Commons get managed, successfully, more often than the fable admits.

Cod, though, was managed, at least on paper. DFO held the licensing power and set every quota. It decided who could fish and with what gear, down to the season. That's exactly the apparatus Hardin's fable says should prevent a collapse, and it existed here in full, and the collapse happened anyway.

A geographer named Patricia Marchak looked at this exact case and argued it isn't really a Hardin-style commons at all, but a tragedy of state mismanagement, since the fishermen never held the management rights; the state held them, and used them badly. I think she's onto something real — more than I gave her credit for the first time I wrote this passage, when I called her point and Hardin's the same structure, one floor up. That was too tidy. Two mechanisms compounded here, and they're worth taking apart separately.

The water ran on Hardin's arithmetic almost exactly: many boats, one stock, and a captain's math that worked out fine on its own no matter what anyone else did. Add a set, run another mile of net, and the fish you don't take, somebody else will — so nobody's restraint would have saved the stock, and nobody had much reason to supply any. That's the offshore fleet, expanding into water the foreign boats had just left, each company running boats it was, from where it sat, entirely reasonable to run.

The ministry ran on different arithmetic entirely. No rival department was racing DFO for the same fish; no second agency's restraint would have gone to waste the moment DFO held back. What DFO had instead was a single actor's problem, repeated every year: cut the quota now, and hand four hundred communities an immediate, front-page catastrophe with a date on it. Leave it uncut, and defer a loss that wouldn't arrive for years, wouldn't carry anyone's name when it did, and wouldn't cost a single election, because the bill always came due on somebody

else's calendar. DFO was racing its own next budget cycle, year after year, and the next one always won.

Two mechanisms, then, compounding rather than repeating. The first — the race on the water — is most of the reason the fish got overfished. The second — the ministry's arithmetic — is most of the reason nobody with the power to stop it ever did — the concentrated cost of cutting kept looking worse than the diffuse cost of waiting, until there was no more waiting left to do. I'm going to keep finding both of these, sometimes on their own and sometimes stacked exactly like this, for the rest of this book.

What actually happened doesn't change under either label: a body that was supposed to be managing this resource used a broken measuring stick, got told the stick was broken, and kept fishing anyway for three more years. If you want to know whether that still bothers me, it does, a little, thirty years on, at a department and a decision I had no hand in and can't do anything about now. There's no tidy place to put that feeling. It's just there.

Every choice in that chain makes sense from inside it, and I want to walk through why before you decide I'm being generous. Keeping the inshore boats fishing made sense — it was their whole economy, and they weren't the ones causing the collapse. Expanding the offshore fleet into water the foreign boats had just vacated made just as much sense from inside a boardroom. So did trusting a catch-based model, right up until there was a specific, hard-won reason not to.

Crosbie is the harder case, and the uglier one. He was a Newfoundlander, the kind of local face Ottawa doesn't usually send up to do its dirty work. Cutting the quota meant handing four hundred communities an immediate, front-page catastrophe. In exchange for what? A fish population that might recover in a decade. It might not. Either way it wouldn't show up as a win for

anyone in office. He had to pay the entire local cost of a diffuse future benefit, on a Tuesday, with his name on it. Almost nobody does that voluntarily. Even knowing what we know now, I'm not sure I'd have cut the quota in his chair, at that podium, with that crowd in front of me. I'd like to think so. I'm not sure.

The stock fell to under five percent of its already-depleted 1990 level within two years. The roughly thirty thousand fishers and plant workers counted as directly out of work is the number you'll usually see quoted; tallies of the wider economic damage, everyone whose income depended on the fishery indirectly, run meaningfully higher. The "temporary" moratorium didn't meaningfully lift until 2024. Thirty-two years, for a two-year fix.

That's the structure I keep bringing up at dinner parties, the ones where you're already regretting inviting me. Individually reasonable, collectively ruinous. Nobody in it was confused about where it was headed. It happened anyway, in slow motion, in front of everybody, for fifteen years.

You'll ask, because you always do, whether this makes me want to ban the whole practice, put someone in charge who can override the fleet and the ministers and the models. I haven't worked that part out yet. It's a question for later in this book, not this letter. I'd rather admit the gap now than pretend the two cases I'm about to give you settle it.

Okay.

I'm out on the porch with this part of the letter now, one lamp on behind me, a moth already finding it. It's 2017. Amazon announces it's opening a second headquarters and invites North American cities to bid. Two hundred and thirty-eight of them do. If you'd asked any single mayor in that pile whether the odds favored their city, most of them would have said probably not, and bid anyway, because probably not beats definitely not, and

definitely not is what not bidding guaranteed. Within days a congressman named Ro Khanna is calling the whole contest a “race to the bottom,” publicly, before the worst offers had even landed. Cities kept bidding anyway, including some whose own mayors would go on to use that exact phrase about their own behavior. Chicago put together something like two billion dollars in tax breaks and free land, by the contemporaneous business press’s tallies, while the race-to-the-bottom framing was already public. Already on the news. Already being said by the people running the bids. New York eventually offered somewhere around a billion and a half. Virginia won with roughly \$573 million and quietly kept the whole thing.

Chicago’s bid came out of Mayor Rahm Emanuel’s office. Boston’s came out of Mayor Marty Walsh’s. Khanna wasn’t just calling the contest ugly from the sidelines, either; he’d also asked, publicly, why the twenty-odd shortlisted cities didn’t band together and negotiate as a bloc instead of racing each other down. Collectively bargain, was the phrase he used, instead of collectively beg. Nobody took him up on it. A joint front only works if every city trusts every other city not to break ranks the moment Amazon calls back, and nobody had a way to guarantee that, so each city kept doing, alone, the thing that made sense alone.

New York’s site collapsed in February 2019, after local political blowback got loud enough that Amazon walked rather than fight it out. Alexandria Ocasio-Cortez, the congresswoman whose district bordered the proposed site, was one of the loudest voices in that opposition. Half the announced prize, gone, after the city had already committed the subsidy on paper.

Economists had been warning about exactly this kind of competition for years before Amazon published its list of criteria.

Not all of them agree on what the warning means — there's a real, unresolved argument among economists about whether bidding wars like this destroy value or just move it from a government's ledger to a company's, and I'm less sure than I'd like to be about which side has the better of it. Whether New York actually came out ahead by walking away is its own contested question, not the clean escape it looks like from the outside. None of that changes the part I'm using the case for: the warnings were public, the bidding didn't slow down, and I don't think it was ever going to.

You've felt a smaller, tamer version of this shape yourself, probably, the first time somebody stood up at a concert and everyone behind them had to stand too, just to see.

When I told you the outline of this on the phone a few weeks ago, you said it in about four words: "This is just greed." Cities wanting the jobs, executives wanting the discount, nobody forced to bid. I want to give that its full weight — it's not a dumb objection, and waving it away fast would be its own kind of dishonesty. Greed is real. Some of the people at those negotiating tables were almost certainly angling for a bigger win than their city needed, and a few of them knew exactly what they were doing. That's part of the picture. But greed doesn't explain the detail that breaks the case open. These people said, on the record, that they knew exactly where this led, and kept doing it anyway. A mayor doesn't warn the public about a race to the bottom and then lose his own city's bid by accident. A governor, Montana's Steve Bullock, used almost the identical phrase a couple of years later, about this same kind of bidding war. He said: "It really does become a race to the bottom." He knew exactly what he was describing while he was describing it. Unilaterally dropping out doesn't stop the race. It just means your city loses for sure instead of maybe.

The inshore fishermen weren't being unreasonable. Neither were the scientists. Crosbie, at his podium, maybe least of all. Add the mayors to that list too, even the ones who'd used the phrase "race to the bottom" about their own conduct and kept typing numbers into the offer sheet anyway. Every one of them was doing something a reasonable person in their seat would have done. They're all climbing the same ladder, and every rung looks sane from where the person standing on it is standing. It's only from above the ladder, looking down at the whole thing at once, that the climb looks like a trap instead of a series of sensible steps.

There's an economist, Thomas Schelling, who spent a whole book on more or less this gap. It's the difference between micromotives, the individually sensible reasons each person has for what they do, and macrobehavior, the aggregate pattern all those sensible reasons add up to, which is sometimes a catastrophe nobody chose and everybody produced. He wasn't writing about fish or tax incentives. He was writing about segregated neighborhoods and traffic patterns, and about the way a lot of small, ordinary preferences can tip into an extreme collective outcome once enough people hold them.

He also pointed out that the same mechanics run the other way, toward things that are stable and genuinely fine. Daylight saving time. The standard layout of a keyboard. A whole category of arrangements everybody follows only because everybody else does, and nobody minds. Worth remembering, before this chapter starts sounding like it thinks every emergent pattern is a trap. Most aren't. These two were.

They're not quite the same machine, even though they rhyme. A commons breaks because the cost of holding back falls on you alone while the benefit, if it ever arrives, is shared among everyone and arrives late. A race to the bottom breaks for close to

the opposite reason: stopping unilaterally doesn't slow anything down, it just guarantees that you're the one who loses, so nobody stops. Different failure. Same result, which is why they get confused for each other constantly, including by me on a bad night.

The first structure has a name: the tragedy of the commons, even with the asterisk its own author later taped onto it. The second has a name too: a race to the bottom. There's a third shape in this family I'm not going to draw out here, something involving a stag and a rabbit, and how much you're willing to trust the guy standing next to you. I could hand you five of these tonight and you'd remember none of them by breakfast, so you're getting two, done right. And the third is the one shape in this whole family that doesn't end in ruin.

What none of this tells you, Nate, is who deserves the anger, and I've stopped expecting a single answer. It's later now, the porch light drawing more moths than just the one by this point, and I think the honest answer has two parts, the same two I walked you through back at the wharf. Some of the anger belongs down at the level of the boats — not at any one captain, but at a shape where nobody's restraint would have saved the stock, so nobody had much reason to supply any. And some of it belongs one level up, at a shape that was never about boats at all: a department that could see the cost of caution clearly and the cost of delay only vaguely, so it kept paying the vague cost, year after year, because the vague cost never came due on anyone's calendar. I don't know how you'd weigh those two against each other in tonnes of blame, and I don't think anyone honestly does. Two different traps. Both real. Both still running, in other rooms, right now.

I think about that crowd on the wharf sometimes, the ones yelling at Crosbie, whenever I catch myself doing the same thing in some other argument entirely: find whoever's standing closest, yell at him, because he's the one who showed up. Blame finds the lowest point in a room the way water does, whether or not anybody built it a drain, and Crosbie was just standing at the lowest point that morning. He didn't take the fish out of the water either.

Tell me I'm wrong about something. You usually can. You will, and I'll probably end up agreeing with about a third of it once I've had a night to sit with it, which is roughly my average with you. And next time I'll tell you about the one structure in this family that doesn't need anybody, anywhere, to be good at all.

2 · The Shadow of the Future

I promised you something at the close of the last chapter: one arrangement in this whole business that doesn't need a single person in it to be good, not the fishermen, not the minister, nobody. Here it is.

In the late 1970s a political scientist at the University of Michigan named Robert Axelrod set out to answer a question he later put plainly. When should a person cooperate, and when should a person be selfish, in an interaction that doesn't end after one round? He didn't answer it from an armchair. He wrote to game theorists and asked them to send in computer programs — entries in what he called a Computer Prisoner's Dilemma Tournament, structured much like a chess tournament, except what the machines played wasn't chess. Entries traveled the way programs still traveled at the end of the seventies: decks of punch cards and typed listings riding the postal service rather than a wire. Richard Dawkins, the evolutionary biologist, would eventually write the foreword to a revised edition of the book that came out of all this, though that's decades ahead of where the story actually starts.

Fourteen people did. They came from five disciplines: psychology, economics, political science, mathematics, sociology, ranging from professors with named coauthors to at least one

entrant who asked Axelrod not to publish their name at all. Each program was a strategy for the iterated prisoner's dilemma. Two players, two hundred moves, a payoff that rewarded mutual cooperation modestly, punished mutual defection more, and rewarded betraying a cooperator most of all. Axelrod ran every entry against every other entry, against a copy of itself, and against a fifteenth non-strategy called RANDOM, five times over, on a university mainframe that handed the results back as a stack of printout, not a screen. By the time the printer stopped, the machines had made nearly a quarter of a million separate choices.

Some of the entries were intricate. Leslie Downing, a psychologist, submitted a program that tried to build a running model of its opponent and calculate, move by move, whichever choice would maximize the payoff. It opened by assuming the worst of the other player and defected at the outset, and the assumption proved self-fulfilling. Nicholas Tideman and Paula Chieruzzi's economics entry ran forty-one lines of code. William Stein and Amnon Rapoport's ran fifty. The entry whose author had asked to stay anonymous ran seventy-seven lines and finished second to last, ahead of nothing but the random-noise control.

Anatol Rapoport, a professor of psychology at the University of Toronto, sent in four lines. His program cooperated on the first move. After that, it did whatever the other player had done to it the move before. Nothing else. No memory beyond the last move, no model of the opponent, no attempt to calculate anything at all. Of the fourteen entries submitted, it was the shortest by a wide margin.

It won.

Axelrod published the results, along with the full text of every program that had been entered. Some time later he ran the tournament again. This time sixty-two entries arrived in the mail

again, from six countries, submitted by people ranging from a ten-year-old computer hobbyist to professors of computer science and evolutionary biology. Every one of them had access to the published report of the first tournament. Every one of them knew, going in, exactly which program had won and exactly how it worked, four lines and all. The rules allowed anyone to submit anyone else's program, credited or not. A lot of the new entries were built by people who assumed the four lines had only won the first time by luck, and set out to construct something that would exploit its willingness to forgive. Every one of those attempts scored worse than TIT FOR TAT itself, in the same field, under the same rules. Cleverness aimed specifically at beating the four lines kept losing to the four lines.

Only one person submitted the same four lines again. It was Rapoport, who had submitted them the first time.

It won again.

One more piece of that story is worth telling before I move on — quickly, because it isn't the part I actually sat down to write to you about. Axelrod didn't stop at two tournaments. He built a third experiment, an ecological one, where each program's success bought it more copies of itself in a simulated next generation, exactly the way a well-fed animal gets more offspring than a starving one. By the fiftieth simulated generation, the bottom third of the original field had all but vanished. One exploitative rule, nicknamed HARRINGTON, did well early by preying on the weak programs around it. Then the weak programs it was living off of went extinct, and it had nothing left to eat. By the thousandth generation it was gone too, as extinct as its prey. That's the whole ecological argument in a sentence: a strategy that only works by feeding on suckers dies the moment it runs out of suckers. Nice strategies, meanwhile, kept growing, generation

after generation, not because they were owed anything by the universe but because they were still finding partners willing to cooperate back. That's what the tournament itself tells you: nice finishes first. The ecological run tells you something a little stronger: nice keeps winning, for as long as you keep playing.

Marisol, I still haven't been able to put down something you told me outside courtroom 4B last month. Twenty minutes before a hearing, eating a granola bar you'd apparently found in a coat pocket, you mentioned that the ADA you've been sparring with for six years had brought you coffee that morning, because you'd both had the same miserable week. You said it like you'd been caught doing something a little embarrassing. I want to try to talk you out of the embarrassment. Not out of the coffee.

Here's the claim, plainly, before I earn it: cooperation doesn't require anyone to be good. It requires the expectation of meeting again. That expectation isn't a fact about human nature you either have or don't. It's something you can build, on purpose, with people who have every reason to hate you, and the tournament above is the best evidence I know of that. A related shape lives in this same family of games: game theorists call it the stag hunt. Two hunters have to decide whether to trust each other enough to chase something big together, instead of settling separately for something smaller and safer. It runs on trust, not incentive — a different engine entirely.

What was doing the work in Axelrod's tournament, in his own accounting, was a bundle of four properties, and they only make sense together. TIT FOR TAT was nice: never the first to defect, which kept it out of fights that didn't need to happen (p. 33). Niceness alone gets you eaten, so it was also retaliatory, punishing an unprovoked defection immediately, no waiting, no strategic patience (p. 44). Retaliation alone traps you in a feud forever, so it

was forgiving too. One punishment, then back to cooperating, letting the other player earn its way back rather than paying for one bad move indefinitely (p. 36). A rule called FRIEDMAN ran the opposite policy, permanent retaliation, never letting go of a single defection — and it scored worse for it, in that same field, under those same rules (p. 36). And none of it matters if the other side can't tell what you're doing. TIT FOR TAT was clear — legible, easy to notice, easy to trust once recognized. Axelrod gave that last property the last word: "its clarity makes it intelligible to the other player, thereby eliciting long-term cooperation" (p. 54).

Read that again as a description of a working relationship instead of a subroutine and it stops sounding like computer science. There's nothing big or dramatic in any of this, which might be exactly the point.

Downing's program was the smartest thing in the room by any ordinary measure, and it isn't close. It built a model of its opponent. It calculated. It tried to reason its way to the correct move instead of just reacting like some kind of thermostat, which is a reasonable way to describe what TIT FOR TAT actually is. Downing's entry came in ahead of only a handful of programs, because reasoning its way to the correct move meant reasoning its way into assuming bad faith early, and the assumption became true the moment it acted on it. Four lines beat all of that. Not because simple is a virtue in itself; I don't think it generally is. Simple happened to be the shape that let the other player see what it was dealing with fast enough to cooperate back, and being seen turned out to matter more than being smart.

I keep thinking about how that would have felt from inside Downing's chair. Modeling the other side is usually good advice, in a courtroom or anywhere else; the trouble is what happens when the model gets too good, too fast. The smarter it got, the

earlier it started expecting the worst, and expecting the worst is close to guaranteeing it. Rapoport's four lines never had an opinion about the other player's character at all. It just answered the last move, and let the record speak for itself.

Axelrod has a phrase for the mechanism underneath all of it: the shadow of the future. Cooperation holds when the future looms large enough over the present that it's worth eating a short-term loss to protect it (pp. 124–126). Shrink that shadow, make the next meeting unlikely, and the same two players who cooperated for years will defect on their way out the door. The shadow isn't fixed, though. You can lengthen it on purpose. Axelrod's own example is almost embarrassingly ordinary: a wedding is a public ritual whose entire function is to make a relationship harder to walk away from (p. 129). A courthouse does the same job by accident. You end up in front of the same judge, the same ADA, in a building too small to let anyone disappear into anonymity, year after year, whether either of you planned it that way or not.

You, Marisol, are what the shadow of the future looks like wearing a person's face. You're not more virtuous than a stranger passing through. You just can't get away from each other. None of this is warm, sentimental stuff. It's arithmetic with a memory. I think about this every time you complain about that particular courthouse, the one with the elevator that only sometimes works and the vending machine that's been out of everything but the diet soda since spring. You've told me you hate the building. I don't think you'd hate a rotating cast of strangers any less. No reputation attached to anybody, no reason for either of you to be anything other than as ruthless as the job technically allows. You once told me the worst opposing counsel you ever had was somebody parachuted in for a single hearing, gone by the afternoon, no incentive to be anything other than as brutal as the

file allowed, because there'd be no next time to answer for it. That's the shadow of the future missing entirely, not as an abstraction but as a specific bad afternoon in a specific room.

This part of the story reaches me through Axelrod's own chapter, which leans on the sociologist Tony Ashworth's research into these trenches rather than the diaries themselves — so this letter is secondhand twice over. Axelrod treats it, formally, as an iterated prisoner's dilemma played not between individuals but between small units, battalions facing each other across a few hundred yards of mud, for months at a stretch (pp. 73, 75). The structure underneath is the same one from the tournament. Defect and you win big for a day. Both sides defect and everyone loses slowly. Both sides hold back and the slow loss is smaller for both. Men trying to kill each other on the Western Front built, independently, wherever units sat still long enough, without anyone ordering it, a working system of not killing each other. Axelrod calls the system endemic to trench warfare — it turned up in the records of nearly every British division — though it lived mostly in the quiet, static stretches of the line and the lulls between big offensives, and the offensives themselves kept destroying it.

It started small, and it climbed. Ration parties get careless because the guns go quiet at the same hour every night, and the quiet just continues (pp. 77–78). Direct fraternization, men climbing out to talk, trade tobacco, shake hands across the wire, got suppressed fast, court-martialed out of a soldier the first few times anyone tried it (p. 78). What survived wasn't the friendly version. It was the quieter one: nobody talking, nobody meeting, just two sides independently deciding not to aim quite as well as they could. Foul weather produced its own truces on top of that, nobody ordering those either, just enough mud and enough cold

that killing the men across the way stopped feeling urgent for an afternoon (p. 78). Life in those trenches was awful by any account, and also, somehow, ritualized down to the minute. None of these three starting points needed a treaty, a handshake, or anybody's permission. They needed two sides that were going to keep facing each other tomorrow, and the next day, and the day after that, for as long as the war lasted, which at the time nobody could say.

The system needed the other side to believe restraint wasn't weakness, so both sides spent real ammunition proving it wasn't. German snipers would put a string of shots into a cottage wall just to prove they could. Not because they wanted to hurt anyone. Because a side that can't demonstrate it's dangerous can't be trusted when it says it's holding back (p. 79). The British artillery ran the same ritual with shells, so regular that German troops started calling it something close to "the evening gun." You could set a watch by it. Men climbed up out of the trench a little before seven some evenings just to watch it land exactly where it always landed, then climbed back down (pp. 79–80, 86).

New units rotating in got briefed by the ones rotating out, not officially, nobody wrote it down. Just which stretch of wire was quiet at which hour, and which regiment across the way could be trusted to keep it that way (pp. 80–81). That handoff is the part I find myself circling back to more than the artillery. Something that has to be re-taught, quietly, unit by unit, every time the personnel changes, and still holds, isn't luck. Ashworth's own phrase for what the ritual was doing, quoted by Axelrod: it served "both sentiments of fellow-feelings, and beliefs that the enemy was a fellow sufferer" (Ashworth 1980, p. 144; Axelrod, p. 87). All three high commands, British, French, German, hated this and tried repeatedly to stamp it out. They wanted men who'd kill efficiently,

not men keeping a private truce with the enemy (p. 81), which still irritates me slightly, uselessly, this many decades on.

One scene from inside it I can't shake. British troops jeered at a German salvo that did no damage. A German soldier, in range, stood up on his own parapet and shouted an apology: "We are very sorry about that; we hope no one was hurt. It is not our fault, it is that damned Prussian artillery" (pp. 84–85). Axelrod's own gloss is that this goes past strategy into something closer to moral regret. I think he's right, and I don't think it undoes anything I've told you. It just means bare incentive and real feeling can occupy the same soldier at the same moment — one doesn't have to have grown out of the other.

Predictability, not decency, was what the arrangement needed from either side.

On purpose, at real cost, because predictable was the only thing the whole arrangement needed anybody to be.

It ended from above, by people who couldn't see what they were breaking. Headquarters couldn't tell whether ordinary shelling was real aggression or theater. So they invented something they could verify: the raid, ten to two hundred men sent to kill or capture in the enemy's own trench, producing prisoners or bodies that couldn't be faked (p. 82) — an unreasonable demand to make of men who'd spent months proving they could do exactly the opposite. You cannot fake a raid the way you can fake a near miss. Once raids were ordered and monitored from headquarters instead of managed locally, nobody nearby could quietly de-escalate the retaliation the way they'd de-escalated everything else for years. One order removed what had built the whole arrangement: repeated contact between people free to adjust to each other.

Axelrod's own conclusion, at the end of that chapter, doesn't try to make the case warmer than it is. His point runs closer to the opposite: that friendship turned out not to be necessary to get reciprocal cooperation going at all, that under the right conditions even people trying to kill each other manage it (p. 87). A version of this story, with a Christmas football match in no-man's-land, gets told constantly; it isn't in Axelrod's account, so I'm leaving it out. What's actually here is stranger than the football story anyway, and it doesn't need a match nobody can quite verify happened.

You asked me something like this once, in a different context: is this just self-interest with a longer memory? It isn't a dumb question, and I want to answer it straight instead of talking around it, because you'd smell it if I dodged. The game theory shows why cooperative behavior is stable, why niceness and forgiveness and legibility survive and spread even among people who owe each other nothing. It does not show that any of it is right. It explains the mechanics of a fact. It doesn't tell you the fact is good. I'm confident about the first part. That stability is real, verified twice, in public, by strangers trying deliberately to break it. I'm far less sure about the second, and a later part of this book exists mostly to climb into that gap rather than paper over it. It would be tidier to tell you the mechanism and the morality turn out to be one and the same, described two different ways. I don't actually believe that.

TIT FOR TAT has a flaw, and it isn't small. Axelrod's tournaments ran on perfect information, no static, no misread signals. Real life has noise: you send a cooperative move and it arrives looking like a defection, or the other side just makes a mistake. Axelrod himself had already noticed a milder version of the problem in the original tournament data, a program called JOSS that got itself caught in exactly this kind of echoing loop

(p. 37). Under noise, plain TIT FOR TAT is fragile: one accidental defection can set off a longer echo of retaliation that neither side meant to start. Martin Nowak and Karl Sigmund picked this up after Axelrod and published the fix in 1992. It isn't retaliating more carefully. It's a fixed, small, calibrated dose of unearned forgiveness: cooperate anyway, sometimes, after a defection, even when you can't tell whether it was deliberate — the forgiving property, taken more seriously than the original tournament had reason to take it. The same two researchers found a further rule in 1993, nicknamed Pavlov, that can beat TIT FOR TAT outright in some conditions by simply repeating whatever move worked last time and switching after a move that didn't. That doesn't undo the earlier result. It's a dispute about which rule wins inside a repeated game, not whether repetition itself is what's doing the work, and that second question is the one this whole chapter has actually been about. The generosity result is what actually matters here.

The same logic that makes TIT FOR TAT need a little generosity is the logic running underneath your courthouse, as far as I can tell. I think about the trench system and your courthouse as two versions of the same machine, one built out of shells and one built out of docket numbers. Both of them ran on the same fuel: neither side could leave, and both sides knew it, and knowing it did more work than any amount of goodwill could have. Take away the guaranteed rematch, in a trench or a courthouse, and the whole arrangement stops making sense to run. Swap the personnel out completely, permanently, no rotation and no next posting, and I'd expect the truce to go with them; the whole mechanism depends on somebody being there next week to remember what happened this week. Call that inference rather than transcription: it's what

the logic underneath the record predicts, and nothing in the record pushes back against it.

I ran into a cruder version of that same forgiveness problem in my twenties, in a library carrel with my jacket still zipped because the heating never reached the back row. A friend had wronged me once, genuinely, not a slight I was inventing to feel important, an actual betrayal, the kind where the facts aren't in dispute even now. I decided the correct response was permanent retaliation: cut contact, hold the grudge, let one bad move define the whole relationship going forward. I told myself this was principled. And it worked, as a story, right up until it stopped working as anything else.

I even had a theory for it, half-remembered from a class, about how you shouldn't reward defection. What I actually was, underneath the theory, was unforgiving, and I dressed it up as strategy because strategy sounded better than what it was. I hadn't finished the book. If I had, I would have run into Axelrod's own numbers telling me flatly that permanent retaliation finished dead last among the rules that never struck first — that grudge-holding was the worst thing a decent strategy could add to itself. FRIEDMAN, the totally unforgiving rule, never found its way back out of a single feud it started. I was running FRIEDMAN on a person I'd known since I was nineteen and calling it integrity. It took about six years and one mutual funeral to even start speaking to him again. I don't know whether the six years bought anything or just cost both of us six years, which is a weird thing to still not know, this many years later. I've never told you that story before tonight. I'm telling it anyway because the tournament data would have told me I was wrong. It would have taken about the time it takes to get through forty pages. I got to those forty pages years later than I should have.

There's nothing shameful in that coffee, Marisol — you got there without needing six years, a funeral, or forty pages of proof, the way I did. You and that ADA built the exact arrangement fourteen strangers stumbled onto independently, decades apart, in a courthouse instead of a laptop. It's a building that smells like every courthouse I've ever been in, floor wax and stale coffee and somebody's nervous cologne. It works because he learned the trick too, the same way the sniper's opposite number learned it, the same way Rapoport's four lines taught sixty-one strangers a lesson none of them could beat even knowing it was coming. Call it plumbing, if that helps you feel less silly about the granola bar, and don't hand the credit to some theory of human goodness neither of you actually needs.

I'll say the honest version once more, since it's the whole reason I wrote any of this down. You don't need the ADA to be a good person, and he doesn't need you to be one either. You both just need to know you'll be standing in that hallway again next month, and the month after, for as long as either of you keeps the job. That's not small, even if it isn't what you were taught to call virtue.

You're the one who taught me the phrase, months ago, not me: lawyers call it a courtroom workgroup, the handful of people who end up in the same room every week for years — judges and prosecutors and defense counsel alike — until they know each other's tendencies better than they know the law. Here's the piece of it I'd be lying if I left out: that workgroup is just as capable of agreeing on whatever clears the docket as on whatever's fair to the person only passing through it once, and nobody in the room has to be corrupt for that to happen. The same coordination that keeps you and that ADA decent to each other is the coordination that can make the two of you, quietly, less accountable to the defendant

who was never in your repeated game at all. I'm naming that here, as the crack Part Two of this book exists to climb into, not settling it tonight.

What the shadow of the future can't tell you is whether the stability it buys is also fair, and that's the question waiting in Valencia, where a water court has been running itself for five hundred years without anyone we'd call a boss.

10 • Three Gates

On the evening of January 27, 1986, engineers at Morton Thiokol's plant in Brigham City, Utah, gathered around a speakerphone in a windowless conference room. Patched in on the other end were NASA managers at Marshall Space Flight Center in Alabama, among them Lawrence Mulloy and George Hardy. They were preparing for the next morning's launch of the space shuttle Challenger. The forecast for the Cape that night called for temperatures near freezing, colder than any previous shuttle launch. No shuttle had ever flown below 53 degrees Fahrenheit.

Six months earlier, Thiokol seal engineer Roger Boisjoly had written to Robert Lund, the company's vice president of engineering, warning that if the O-ring erosion went unaddressed the company stood to lose a flight and the people on it, and asking that a team be assigned to the problem. The memo went into the file.

On the call that night, Thiokol's engineers — Boisjoly among them, along with Arnie Thompson — laid out O-ring erosion data from earlier flights. They recommended against launching below 53 degrees. It was the floor for every flight the shuttle had ever made. Lawrence Mulloy, NASA's project manager for the solid rocket boosters, pushed back hard. According to later testimony,

he asked Thiokol when it wanted him to launch. The job witnesses remembered him landing: next April?

Thiokol's managers asked to go off-line and caucus privately. They said they needed five minutes. It ran close to thirty. No new engineering data was gathered in that half hour, in that room, on that phone line. According to testimony given afterward, Jerry Mason, Thiokol's senior vice president, turned to Lund and told him to take off his engineering hat and put on his management hat.

Lund changed his vote.

Boisjoly later described the moment this way: "At that moment I felt totally helpless and felt that further argument was fruitless, so I, too, stopped pressing my case." When the caucus ended, four senior Thiokol managers gave NASA a verbal recommendation to launch. The engineers who had argued against it were not asked to vote.

Challenger lifted off the next morning at 11:38. Seventy-three seconds later, it broke apart. All seven crew members were killed.

That's the version everyone tells, and it isn't wrong so much as it's shallow. Diane Vaughan spent years inside the primary documents for her history of the decision, and what she found is colder than one bad night. O-ring erosion had already been observed, discussed, and reclassified as acceptable risk across a long run of earlier flights. Each launch that survived became precedent for the next one. Nobody on that call was breaking a rule they knew they were breaking. They were following a rule that had already drifted, one launch at a time, without anyone experiencing the exception as an exception.

Twenty years later, the same shape of failure showed up at General Motors. An engineer named Ray DeGiorgio approved an ignition switch that didn't meet the company's own torque

specification, and referred to it in an internal memo as “the switch from hell.” In 2005, GM engineers proposed a fix costing under a dollar a unit. The internal review concluded there was no “acceptable business case” for making the change, and logged the defect as a customer-satisfaction issue rather than a safety one. The switch itself was small enough to fit in a shirt pocket. Nobody in that chain was covering up a crime, and nobody was being unreasonable, exactly. Each person was applying a category that, from where they sat, looked defensible: this is a satisfaction complaint, not a safety recall; this is a routine part change, not one that needs a new part number. GM’s own internal investigation, commissioned by its board, later found no evidence of a coordinated conspiracy. That’s what makes it worse, not better. A hundred and twenty-four deaths came out of a sequence of individually reasonable-sounding calls, made by people who each thought they were staying inside their lane.

I have a friend, an emergency physician named Priya, who almost majored in viola before deciding medicine was the more stubborn addiction; the case still sits, unopened most years, in a corner of her apartment. She called me once from a hallway outside a supply closet, near midnight, still in the shoes she’d been standing in for sixteen hours, to tell me about a code she’d just run. She had overridden a family’s hesitation on a DNR conversation. Pushed past it. Made the call herself, because in the moment she was certain that stopping would be the wrong thing to do. She told me: “I knew it was the right call.” I believe her. That’s exactly the problem.

Everyone who has ever done something catastrophic in the name of an exception also knew it was the right call in the moment. Mulloy knew it. The Thiokol managers who traded their engineering hats for management ones knew it, or told themselves

they did, in a room that had been arguing for hours and badly wanted to stop. The feeling of being unmistakably right doesn't sort the people who are from the people about to end up in a commission's transcript. It's what everyone says right before, and right after. Those are the only two times anyone says anything.

The standing rules are the default for a reason, and none of that reasoning depended on anyone's character being good enough. So: you can't trust your own judgment about when you're entitled to break one. I don't think I can. I don't think anyone competent enough to be dangerous can. The fix most people reach for is a better test. A sharper definition of catastrophic. A clearer account of last resort. More rigor in the moment of deciding. I believed that too, until I noticed the moment of deciding is exactly the moment you can't trust. Refining the test aims a better instrument at a broken sensor.

The actual fix has to be a procedure that runs before you're allowed anywhere near the override at all — one that skips *this justified*, because that question is rigged from the inside, and asks something cruder, and earlier, instead.

It's tempting to think there's a shortcut here: skip the machinery, just put better people in the room. Daniel Kahneman's distinction between a fast, intuitive mode of judgment and a slow, deliberate one is useful here, though it isn't flattering to the shortcut. The fast system doesn't get more accurate in people with better values. It gets confident faster. A conscientious person under pressure reaches the wrong call with exactly the same felt certainty as a careless one, which is why character alone isn't sufficient. You need something outside the person too — something that doesn't care how good the person is.

The earlier question is a trigger. It has to be bright-line, almost stupid in how little judgment it requires: is there imminent harm

on an irreversible scale, or has the normal error-correcting machinery itself been disabled? Is there, in other words, no one left upstream to appeal to? Not “this seems really bad.” Not “I have a strong intuition.” A threshold dumb enough to check while furious, while exhausted, while certain — because those are exactly the states in which judgment about subtler questions is worst.

If the trigger doesn’t clear, you don’t get to the interesting part of the conversation. That’s annoying. It’s also the entire point. The trigger works like a check: indifferent to how sure you feel, immune to being argued with.

This is why the trigger has to be a bright line instead of trained judgment. Judgment renormalizes; it drifts a little at a time, the way the risk category drifted at Thiokol, and nobody inside the drift feels it happening. A fixed threshold can’t do that. Fifty-three degrees doesn’t creep upward a degree a year. A channel either failed or it didn’t. That’s the case for building the trigger stupid: a standard too dumb to argue with is also too dumb to drift. But the trigger and the caucus answer two different failures. The years of quiet reclassification are the slow rot a bright line resists. The half hour on the phone, Mason telling Lund to swap hats, sits on top of it: a discrete override laid over a culture that had already renormalized. These gates don’t touch the first failure. They’re built for the second.

If it does clear, the second gate is a delay. This is the part people hate most, and it’s the part I’d fight hardest to keep. There’s a concept from the Carnegie Mellon economist George Loewenstein: the hot-cold empathy gap, an idea he began building in the nineteen-nineties and formally named a decade later. It describes how badly people in an emotionally “hot” state predict what a

“cold” version of themselves would decide. The reverse prediction goes badly too.

You are never colder than in a calm hallway conversation about hypotheticals, the coffee in your hand still warm because nothing is actually happening yet. You are never hotter than ninety seconds into the actual thing. The delay is engineering, crude but real: it puts a cold version of you back in the room before the override executes, because the hot version is provably not a reliable witness to your own reasoning.

Versions of this already exist, scattered and unglamorous, in places nobody thinks of as high moral technology. Elective surgery usually comes with a mandatory cooling-off period between consent and the operating room, precisely because a frightened, freshly diagnosed patient isn’t the same decision-maker as that patient a week later. Airline crews are grounded after a certain number of duty hours regardless of how alert any pilot insists he feels, because feeling alert is exactly the judgment fatigue degrades first. Nobody thinks the waiting period or the crew-rest rule is an insult to anyone’s competence. They’re a plain admission that competence is state-dependent, and the states we’re worst in feel, from inside, most like clarity.

None of this is new machinery, and I’d rather say so than let the chapter imply otherwise. Jon Elster spent a career cataloguing exactly this kind of arrangement: a person binding their future, worse-judgment self in advance, the way Ulysses had his crew lash him to the mast before the sirens started singing. Institutions gate their own exceptions the same way, procedurally, all the time. An institutional review board signs off before a study on human subjects can start. A missile launch takes two officers turning two keys neither can reach alone. A warrant clears a judge’s desk before the search happens, not after. Threshold deontologists have

spent decades on the substantive half of this — how catastrophic the harm has to be before a constraint gives way. What’s actually new here is narrower: taking gating built for institutions and making one person, alone, run it on themselves, plus a review that runs no matter how the story ends.

Then comes the third gate: a review, no matter what you decide or how it turns out. Written down in advance that there will be one. Not contingent on anyone catching you.

The “no matter how it turns out” part is doing more work than it looks like it’s doing. It’s tempting to grade the override by its result: it worked, so the reasoning must have been sound; it went badly, so the reasoning must have been reckless. That’s outcome bias, and it’s exactly backward. A reckless override that lands well teaches you nothing except a false confidence you’ll cash in later, at a worse moment. A careful, well-gated override that still goes badly, because the situation really was as bad as it looked, shouldn’t be punished as if the process had failed. The review has to grade the procedure, not the outcome, or the whole architecture collapses into “do whatever works” — which only sounds like a standard because it’s shaped like a sentence.

Only after all three gates clear does the substance start to matter at all, and even then it doesn’t arrive as a checklist you run down in order. It arrives because each condition, taken alone, has a hole the next one exists to plug — the way a rag wedged into a leak buys time without fixing the wall.

Start with the harm itself: was what you were about to prevent actually catastrophic and irreversible? Priya’s code clears that easily — a stopped heart is about as irreversible as harm gets. Harm alone proves nothing, though, since it might have happened regardless of what you did. So the second condition is attribution: did following the rule cause it, or was the rule just standing

nearby when something else went wrong? For Priya, plausibly yes — the delay her family's hesitation would have cost doesn't reopen. Attribution isn't enough either, because there might have been a gentler path nobody looked hard enough for — which is why last resort is the third test. We talked it through late on the phone, my kitchen the only lit room in the apartment. Priya's honest answer was that she doesn't know. She never went back to check, and I didn't push her to.

Last resort has its own blind spot: it asks you to certify your own thoroughness, and you're the worst possible judge of that — you're inside the hot state, grading your own homework. Articulability is the fix, and it's worth slowing down on, because it does more work than the other four conditions combined. The real question isn't whether you exhausted the alternatives. It's whether you could say what you did out loud to someone with no stake in your being right, and let them adjudicate it instead of nodding along. Not a hypothetical stranger, either — a specific, reachable one, because an imagined stranger will always rule the way you need them to. Priya did the real version on her own, unasked. She called me, someone with no stake in her hospital's org chart, and made herself say the whole thing out loud before deciding what she thought of it.

Even that has a hole. A living, skeptical stranger can still be talked past if your own interest in the outcome is doing the talking without your noticing. That's the fifth condition: has that interest been scrubbed out, or is it quietly steering from underneath? None of this is worth asking until the trigger and the delay have cleared, because a person capable of failing the substance is equally capable of failing to notice they've failed it. That's why the order matters more than any single condition in it.

Run Priya's code back through the gates themselves. The trigger: imminent, irreversible harm, no time to appeal upstream? Plausibly yes — that's what a code is. The delay: a rough approximation of a cold read, a beat before acting? That one I don't know, and neither did she. She wanted to argue about whether her reasons were good enough, and I kept steering her back to whether the room had been built to let her reason at all. That ordering, substance last, is the actual point. Most of the discomfort came from how unsatisfying it is to be told the interesting question isn't the one you were ready to answer.

I described the gates to Priya a few weeks later. She was quiet for a second. Then she said: "This is bureaucracy applied to conscience." She didn't become a doctor to run a checklist past a delay timer before she was allowed to act like a person with functioning moral instincts. I take that seriously rather than wave it off. Not least because I've been on the other end of it myself.

Here's the admission, and it isn't a comfortable one. Twice in the last five years I have used procedural language — *I need more information, I need to think it through properly, I don't want to act rashly* — to describe what was actually a decision I was too scared to make. Once about a job I should have quit eighteen months before I did. Once, worse, about a friendship I let a mutual acquaintance keep damaging, because ending it cleanly would have meant admitting I'd misjudged the person for a decade. Both times, "I'm being rigorous" was a nicer story than "I'm being a coward," and I told myself the nicer one until it stopped being available.

So I understand exactly what Priya's objection is pointing at. A gate can be a shield for genuine restraint. It can also be a shield for the same old avoidance, wearing a lab coat. From the inside, those two feel identical. I don't think the gates in this chapter collapse

into that failure, but I hold that less firmly than everything else here. The vocabulary of “procedure” is exactly the vocabulary a coward would also reach for, and I have reached for it. I think the difference is this: a real gate has a trigger a third party could check without knowing you, and a fake one only has a mood. My stalling on the job, and on the friendship, had no trigger at all. It just had a feeling that lasted eighteen months. That’s the tell. A procedure that never resolves isn’t a procedure. It’s an excuse with a longer runway. It’s the same order problem I keep running into elsewhere in this book — decide first, on instinct, then build the slower justification after — just wearing a lab coat this time instead of a rushed one.

What I’d say back to Priya is that the gates function as a guard against the specific failure mode where instincts are worst — not as a replacement for them — which is precisely when the stakes are highest and the clock is shortest. Those are the same conditions, weirdly, that make instincts feel most trustworthy. The pairing is the mechanism.

There’s a whole further argument I could make here with Milgram’s obedience numbers. I’m leaving it out — Challenger already makes the point, and I don’t want to borrow trouble from a data set with an asterisk on it.

Here’s a harder version of Priya’s objection. It’s the one that worries me: Peter Buxtun.

Buxtun was a venereal-disease investigator with the U.S. Public Health Service when he learned around 1966 what the Tuskegee study actually was. Men in Macon County, Alabama, were being watched for decades while syphilis was allowed to run its course, untreated. Buxtun did the thing you’d want a person to do: he climbed the internal channel instead of going around it. He filed a formal written objection in November 1966. A Public Health

Service panel reviewed it and told him the study wasn't finished yet.

He filed again in 1968, arguing that the men in the study didn't understand what was being withheld from them. The panel reviewed it again, and told him no again — through the review process itself, not some rogue official ignoring the rules. It took a reporter, Jean Heller of the Associated Press, breaking the story in 1972 — six years after Buxtun's first letter — to end it: attrition, an outsider forcing what two internal reviews wouldn't. And I find, honestly, that I'm still a little angry about this, decades later, at a study I had no part in and a panel whose members are almost certainly dead. The study was halted within months of the exposé. A class-action settlement followed in 1974. The scandal directly produced the National Research Act of 1974 and the modern system of institutional review boards, which exist because the informal channel Buxtun climbed twice was, in the end, just a mirror.

My gates, as I've described them, run the deliberation and the documentation and the review through the institution. What happens when the institution is what's captured? Buxtun's objection went to a panel stacked with people whose careers depended on the study continuing. This is a documented failure of the architecture I've just spent pages defending: six years, a captured panel, a body count.

So the fix has to include this, even though it's the part I'd rather not have needed to write. When the trigger is institutional capture, the documentation gets sealed or delayed rather than routed through the captured channel. The deliberation clock that would normally protect you from your hot self doesn't apply here, because the same delay that guards against a rash override is the delay Tuskegee used to bury Buxtun for six years.

Underneath that is something worse. Sometimes the failure runs the other way entirely: complying — filing the correct form, through the correct channel, twice, getting the correct rejection both times, and going home. The gates I've described are built almost entirely to catch the first kind. They need a second set of eyes turned on the second, people complying when the apparatus around them has already gone bad, or the chapter is only teaching half the skill.

Cynthia Cooper is the version where the gate worked. In the summer of 2002, as WorldCom's vice president of internal audit, Cooper and her team found roughly \$3.8 billion in operating expenses improperly booked as capital investment — the kind of accounting fiction that makes a dying company's books look healthy. She did the work quietly, mostly at night, to avoid tipping off the CFO whose numbers she was checking. That's a small, human detail I don't want to lose: auditors working after hours by lamp light, checking books they weren't supposed to be checking yet.

What Cooper did differently from Buxtun was route around her own reporting line, which ran straight toward the people implicated, and take her findings instead to Max Bobbitt, chairman of the audit committee, a body structurally outside management's control. It worked because WorldCom, unlike the Public Health Service reviewing Buxtun, had actually built an alternate channel, and built it with teeth. Cooper was named one of Time's Persons of the Year for 2002, alongside Sherron Watkins of Enron and Coleen Rowley of the FBI. All three women, in the same twelve months, found a channel their institutions had built on paper, and walked through it.

Run Buxtun and Cooper side by side and you get the whole shape of the problem. The same procedural instinct, filed through

two different architectures, produced a cover-up in one case and, in the other, the largest corporate accounting fraud yet uncovered, stopped cold. An audit committee and a review panel are the same kind of object on an org chart. Only one of them turned out to be real when someone pushed on it.

There's an objection sitting right under all of this: deciding whether the trigger has fired is still a judgment call, made by the same person the architecture doesn't trust. Asking "does my situation qualify" instead of "is my override justified" might just be moving the self-deception one door down the hall. I think the honest answer is that the trigger asks a narrower, more checkable question — Buxtun and Cooper are the evidence, not just the illustration, since neither case turned on anyone's inner sense of justification, and both turned on facts a stranger could verify without knowing either of them. That's not nothing. It isn't everything either. A person determined enough to lie to himself can lie about facts almost as easily as feelings. The gates raise the cost of self-deception. They don't retire it.

Which means the gates probably can't be one fixed threshold for everybody, since what a person can appeal to varies by role. A soldier starts with the most scaffolding — a chain of command, and above it a body of law — though it's only as good as the appeal reaching someone outside the failure. A physician like Priya has a hospital ethics board, small enough that a bad one gets noticed by the people who work there every day. Strip away the board, the license, and the chain of command, and what's left is the private citizen, with almost nothing above them — a lawyer, a journalist, their own name attached to their own claim. Add a fourth, the civil servant, who is a case strange enough to get their own paragraph. Of the four, the private citizen holds the thinnest

reed, and that probably means the trigger should sit lower for them, not higher.

The civil servant is the interesting case, and I want to slow down on it rather than filing it next to the other three. On paper, every federal agency has an inspector general's office, built for exactly this: a channel one level outside the chain it's supposed to be checking. In practice it functions in something closer to half of them, with no reliable way to know in advance which half you're in. Maybe the letterhead connects to a person with an actual mandate. Maybe it connects to a windowless office that hasn't issued a finding against its own agency in a decade. A soldier can look up the chain of command and see it. A physician can walk down the hall and check whether the ethics board is real. The civil servant can't test in advance whether the scaffolding is load-bearing steel or a stage flat.

I think the role doesn't just move the trigger. It probably changes which condition is hardest to satisfy too, though I hold this less firmly than anything else here. My best guess: for Priya, self-interest is the easy condition, since nobody profits from a code going one way rather than another in any sense that would survive being said aloud to a stranger. For the civil servant staring at an inspector general's office that may or may not be real, articulability is the hard one — the skeptical third party it asks you to imagine may not exist anywhere reachable. Naming the load-bearing condition for your role is most of the real work.

Here's the part I don't want to smooth over. "My role licenses a lower threshold" is one clause away from "my case is special," which is the exact self-serving move this whole chapter has been trying to build a wall against. I don't have a clean answer for who checks the person setting their own role's threshold, and I'm

aware that's precisely the shape of every piece of special pleading this chapter has warned about, including, possibly, mine.

I'm confident about the first two gates. I'm a lot less sure where the trigger line should sit for someone with nobody upstream at all, and I don't know of anyone who has solved it cleanly.

I don't have this fully worked out, and I'd rather say that plainly than tie it off with something reassuring. I suspect a version of these gates would have slowed down some of the good rule-breakers of history along with all the Mulloys, and I don't have a clean way to make it slow down only the second group. I'll put a real date on a guess, so you can hold me to it: by 2036, at least half of the twenty largest U.S. hospital systems by revenue (take whatever ranking is current that year) will have a published policy for a reporting channel that bypasses the chain of command entirely, sealed rather than routed through the compliance office whose failures it exists to catch. Tuskegee-shaped failures keep recurring in different clothes.

Robert Jackall, who spent years embedded with corporate managers and is bleaker than I am, has a line I think about constantly: "What is right in the corporation is what the guy above you wants from you." I think he's too cynical. I think he undersells how many people inside a hierarchy are quietly building the Cooper version of the channel, not the Buxtun version. But I understand why he landed where he landed, and most days I'm not fully sure he's wrong and I'm right.

I never found out how Priya's patient did.

She didn't say, and I didn't push. Some stories a friend tells you are for the story, not for the ending. You learn that after a while.

She called again a few weeks after that first night, at a decent hour for once, just to tell me she'd started keeping her own log. Nothing official. Nothing anyone at the hospital had asked for. A

private file, a date stamped on every entry, sealed in a drawer that wasn't the chart. Not entirely to protect herself, though it probably did that too. She said it was so the version of her that was sure, the one from the middle of the code, would always have to answer, eventually, to the version of her that hadn't been in the room yet.

She meant something more specific than a record for people who might be wrong, though it took me a second call to hear it that way. She meant a record for people who, in the moment, cannot tell the difference between being right and being sure. Near midnight, coffee gone cold, that's everyone I know. It's me.

Notes

Sourced claims from manuscript/frozen/ch-01.md, in order of appearance. Citations drawn from reviews/ch-01/verify.md (independent re-check, not the Researcher's evidence.md/cases.md packet) except where noted. Four items were PASS-WITH-NOTE in verification and were corrected in the frozen text before freeze (fog/radio sensory detail cut; "closer to double" softened to "meaningfully higher"; Sinclair quote reattributed via Mason; Cabot's report reattributed to the Milanese envoy); the endnotes below cite sources for the claims as they now appear.

1. Confrontation at Bay Bulls, Newfoundland (~30 km south of St. John's), July 1, 1992, crowd of roughly 1,000, at what was billed as a Canada Day event. The Independent, <https://theindependent.ca/commentary/seasplainer/who-took-the-fish-from-the-goddamn-water/> ; MUN Luminus, <https://www.mun.ca/alumni/luminus/online/from-desperation-blooms-beauty/>
2. Crosbie dressed in a navy wool coat and tie, arriving "dressed for Ottawa," among a crowd in ball caps and nylon jackets. The Independent, <https://theindependent.ca/commentary/seasplainer/who-took-the-fish-from-the-goddamn-water/>
3. Crosbie's exact quote: "Why are you yelling at me? I didn't take the fish from the goddamn water, so don't go abusing me." Confirmed verbatim across multiple sources. The Independent (above); NL Proud, <https://www.facebook.com/realNLProud/>

posts/i-didnt-take-the-fish-from-the-goddamn-water-so-dont-go-abusing-metoday-on-the-3/1325661024627035/

4. Follow-up exchange (“who took it then” / “you and your goddamn people took it”). The Independent (above).
5. Moratorium announced July 2, 1992, called “temporary,” originally framed around two years; put roughly 30,000 fishers and plant workers out of work in a single announcement, the largest mass layoff in Canadian history; emptied more than 400 communities with no other major employer. CBC, <https://www.cbc.ca/player/play/video/1.3334884> ; The Independent (above). Note: other tallies cite 19,000–40,000 depending on whether direct or direct-plus- indirect losses are counted (Steele, Andersen & Green, “The managed commercial annihilation of northern cod,” *Newfoundland Studies* 8, no. 1 (1992): 34–68); 30,000 is the standard, most-repeated figure for direct job loss.
6. Moratorium did not lift, even partially, until 2024; thirty-two years for what was announced as a two-year measure. DFO reopened a limited commercial fishery (TAC 18,000 tonnes) June 26, 2024, the first reopening since 1992. SeafoodSource, <https://www.seafoodsource.com/news/supply-trade/canada-reopens-northern-cod-fishery-after-32-year-moratorium> ; CBC, <https://www.cbc.ca/news/canada/newfoundland-labrador/nl-northern-cod-fishery-2024-1.7246735> ; Canada.ca, <https://www.canada.ca/en/fisheries-oceans/news/2024/06/the-government-of-canada-announces-the-historic-return-of-the-commercial-northern-cod-fishery-in-newfoundland-and-labrador.html>
7. The 1497 report of Newfoundland’s cod stocks that reached Europe is not a surviving letter from Cabot himself; it is a December 18, 1497 letter from Raimondo di Soncino, the Milanese envoy in London, to the Duke of Milan, relaying what Cabot’s crew had reported on their return (corrected from an earlier draft’s “Cabot wrote home” — see reviews/ch-01/verify.md item 34). Elfinspell, primary-source text of Soncino’s letter, <https://elfinspell.com/PrimarySourceCabot.html>

8. Peter Sinclair, quoted: overfishing by local inshore fishers “must be discounted as a major factor” in the collapse of the cod stocks, even though they “seem to bear the brunt” of the problems that followed. Verified only via a secondary source quoting Sinclair, not Sinclair’s own original publication directly — per the logged decision in `reviews/ch-01/decisions.md`, cited here via Mason. Sinclair, P. R. (1996), as quoted in Mason, Fred. “The Newfoundland Cod Stock Collapse: A Review and Analysis of Social Factors.” *Electronic Green Journal*, no. 17 (2002). <https://escholarship.org/uc/item/19p7z78s>
9. Before 1977, foreign trawlers were the dominant driver of overfishing, working water Canada did not yet control; after Canada declared its 200-mile exclusive economic zone (effective Jan. 1, 1977) and the foreign fleet was pushed out, Canada’s own offshore dragger fleet expanded rapidly to take up the slack and became the dominant post-1977 driver. Standard scholarly account; Sinclair, P. R. (1996), as synthesized in Mason (2002), above.
10. In 1987 the inshore fleet — through the Newfoundland Inshore Fisheries Association — commissioned and paid for an independent audit of the government’s stock-assessment model, run by American fisheries scientist D. L. Alverson, submitted to Minister Tom Siddon. Confirmed per `reviews/ch-01/verify.md` item 14; see also Alverson, D. L. et al., *A Study of Trends of Cod Stocks off Newfoundland and Factors Influencing Their Abundance and Availability to the Inshore Fishery* (1987).
11. Alverson, quoted: “It took several years to find it, and by then, the stocks were on the downturn. I think it came as a shock to all of the fishermen and some of the scientists.” Confirmed verbatim as cited in Harris, Michael. *Lament for an Ocean: The Collapse of the Atlantic Cod Fishery*. Toronto: McClelland & Stewart, 1998, p. 104.
12. By 1989 CAFSAC’s own scientists were recommending catch cuts of more than half; the government cut only about 15 percent. Confirmed per `reviews/ch-01/verify.md` item 16.

13. The 1992 quota Crosbie set, 187,969 tonnes, exceeded the entire industry's catch the previous year, 129,033 tonnes (rounded in this audio script, per the logged decision in `reviews/ch-01/decisions.md`, to "just under one hundred ninety thousand tonnes... about one hundred thirty thousand"; exact figures are preserved in the print manuscript). Confirmed exactly per `reviews/ch-01/verify.md` item 17, multiple independent sources.
14. Alan Finlayson's finding that the inshore fleet's warnings were filtered out before reaching policymakers, and that when doubt was raised in the early 1980s and again when the Alverson report was tabled in 1987, government decision-makers ignored or quieted the naysayers. Finlayson, A. C. *Fishing for Truth: A Sociological Analysis of Northern Cod Stock Assessments from 1977–1990*. St. John's: Memorial University of Newfoundland, Institute of Social and Economic Research, 1994; as synthesized in Mason (2002), above. Not independently re-checked against the Finlayson primary text by the Gate 4 Verifier — see `reviews/ch-01/verify.md`, which does not carry a separate line item for this claim — but corroborated as internally consistent with the rest of the research packet in `reviews/ch-01/cold-read.md`.
15. Garrett Hardin, "The Tragedy of the Commons." *Science* 162, no. 3859 (1968): 1243–48. The herdsman/pasture example is explicitly the essay's "warm-up" for its actual argument about human population control.
16. Hardin borrowed the herdsman example from William Forster Lloyd, an Oxford economist who used it in lectures in 1833, and credits Lloyd by name in the 1968 essay. Lloyd, William Forster. *Two Lectures on the Checks to Population*. Oxford, 1833.
17. Hardin, quoted: "the weightiest mistake in my synthesizing paper was the omission of the modifying adjective 'unmanaged.'" Hardin, Garrett. "Extensions of 'The Tragedy of the Commons.'" *Science* 280, no. 5364 (1998): 682–83. Confirmed verbatim against the primary text. <https://www.science.org/doi/10.1126/>

science.280.5364.682 ; full text, http://www.pelagicos.net/MARS6910_spring2013/readings/Hardin1998.pdf

18. Patricia Marchak's "tragedy of state mismanagement" framing — the argument that Newfoundland cod was not a true commons, since fishers never held management rights and the state (DFO) did. Confirmed as Marchak's own coinage. Marchak, Patricia. "Uncommon Property." In *Uncommon Property: The Fishing and Fish-Processing Industries in British Columbia*, edited by Patricia Marchak, Neil Guppy, and John McMullan. Toronto: Methuen, 1987.
19. 238 cities and regions submitted bids for Amazon's second headquarters (HQ2). Confirmed per [reviews/ch-01/verify.md](#) item 23.
20. Ro Khanna publicly called the HQ2 contest a "race to the bottom" within days of Amazon's September 7, 2017 announcement. Confirmed per [reviews/ch-01/verify.md](#) item 27 (contemporaneous reporting, GeekWire and The Intercept; no single stable URL retained during verification).
21. Chicago's incentive package, "something like two billion dollars," by contemporaneous business-press tallies (\$2.0–2.25B range). Confirmed per [reviews/ch-01/verify.md](#) item 24; business-press sourced (Fortune, Axios, GeekWire), no single stable URL retained.
22. New York's offer, "roughly one point five billion dollars" (\$1.525 billion in direct incentives). Confirmed per [reviews/ch-01/verify.md](#) item 25.
23. Virginia's winning package for the "National Landing" site (Arlington/Alexandria), roughly \$573 million. Confirmed per [reviews/ch-01/verify.md](#) item 26.
24. Chicago's bid came from Mayor Rahm Emanuel's office; Boston's from Mayor Marty Walsh's. Confirmed per [reviews/ch-01/verify.md](#) item 31.

25. Khanna's "collectively bargain instead of collectively beg" proposal, urging the ~20 shortlisted cities to negotiate as a bloc — the title/ framing of Khanna's January 2018 piece. Khanna, Ro. "Why Don't 20 Cities on Amazon's HQ2 Shortlist Collectively Bargain Instead of Collectively Beg?" The Intercept, Jan. 22, 2018, <https://theintercept.com/2018/01/22/amazon-headquarters-shortlist-hq2/> ; reposted, <https://khanna.house.gov/media/in-the-news/why-dont-20-cities-amazons-hq2-shortlist-collectively-bargain-instead-collectively>
26. New York's site collapsed February 14, 2019, after local political opposition; Alexandria Ocasio-Cortez, whose district bordered the proposed site, was one of the most prominent opponents. Confirmed per reviews/ch-01/verify.md items 29–30.
27. Montana Governor Steve Bullock, quoted: "It really does become a race to the bottom," used of interstate incentive competition roughly two and a half years after Khanna's remark. The Hill, Feb. 17, 2020. Confirmed verbatim per reviews/ch-01/verify.md item 33; no stable URL retained during verification.
28. The wider economic toll of the cod moratorium — beyond the ~30,000 directly counted — running "meaningfully higher" (softened from an earlier draft's unpinned "closer to double," per the Gate 4 fix in reviews/ch-01/decisions.md). Steele, Andersen & Green (1992), above, citing roughly 19,000 direct plus up to 20,000 further indirect job losses.

Total: 28 endnotes.

Note: the fog-and-radio sensory detail at Bay Bulls (PASS-WITH-NOTE, reviews/ch-01/verify.md item 7, unverifiable from any accessible source) was cut from the frozen manuscript at Gate 4 and does not appear in this audio script; it is not endnoted here. The Nate correspondence device and the author's personal reflections (the driveway argument, the porch, the "I wanted it to be the fishermen" admission) are not drawn from the evidence packet and are not

endnoted, consistent with the treatment of the prologue's Sarah material and Chapter Ten's Priya material.

Sourced claims from `manuscript/frozen/ch-10.md`, in order of appearance. Citations drawn from `reviews/ch-10/verify.md` (independent re-check, not the Researcher's `evidence.md/cases.md` packet). Three claims failed initial verification and were corrected in the frozen text before freeze; the endnotes below cite sources for the corrected claims as they now appear.

1. Teleconference on the evening of January 27, 1986, Thiokol engineers and NASA managers (including Lawrence Mulloy and George Hardy) discussing forecast cold temperatures ahead of the January 28 Challenger launch. Rogers Commission Report, Vol. I. <https://www.nasa.gov/history/rogersrep/v1p247b.htm>
2. No shuttle had ever flown below 53°F; Thiokol's no-launch-below-53°F engineering recommendation. Rogers Commission Report. https://en.wikipedia.org/wiki/Rogers_Commission_Report
3. Boisjoly's July 31, 1985 memo to Robert Lund warning that unaddressed O-ring erosion put the company in jeopardy of losing a flight and the people on it (corrected from an earlier draft that conflated this memo with a "board of inquiry" line actually spoken by Thiokol's Al McDonald on the January 27, 1986 call — see `reviews/ch-10/verify.md` item 9). Memo text: Online Ethics Center, <https://www.onlineethics.org/cms/12703.aspx> ; full memo reproduced at Letters of Note, <https://lettersofnote.com/2009/10/27/the-result-would-be-a-catastrophe/>
4. Mulloy's "next April?" line, asking Thiokol when it wanted him to launch. Most concrete traceable source: Allan McDonald, *Truth, Lies, and O-Rings* (2009), <https://www.goodreads.com/book/show/5070507-truth-lies-and-o-rings> ; also NPR, "Remembering Roger Boisjoly: He Tried To Stop Shuttle Challenger Launch," <https://www.npr.org/sections/thetwo-way/>

2012/02/06/146490064/remembering-roger-boisjoly-he-tried-to-stop-shuttle-challenger-launch

5. Thiokol's managers requested roughly five minutes for a private caucus and took close to thirty, generating no new engineering data. Houston Public Media, "40 Years After Challenger," <https://www.houstonpublicmedia.org/articles/news/nasa/2026/01/28/541921/40-years-after-challenger/>
6. Jerry Mason telling Robert Lund to take off his engineering hat and put on his management hat. Surfing Complexity, "Putting On a Different Hat," <https://surfingcomplexity.blog/2021/02/26/putting-on-a-different-hat/>
7. Boisjoly quoted: "At that moment I felt totally helpless and felt that further argument was fruitless, so I, too, stopped pressing my case." Online Ethics Center case archive. <https://onlineethics.org/cases/management-decision-overrides-recommendation-not-launch>
8. Four senior Thiokol managers gave NASA a verbal recommendation to launch; the engineers who argued against it were not asked to vote. Rogers Commission Report, corroborated across Houston Public Media and NPR retrospectives (sources above).
9. Challenger lifted off at 11:38 a.m. EST on January 28, 1986; broke apart 73 seconds later; all seven crew members were killed. Rogers Commission Report, Vol. I; NASA History Office. <https://www.nasa.gov/history/rogersrep/v1p247b.htm>
10. Diane Vaughan's finding that O-ring erosion had been observed, discussed, and reclassified as acceptable risk across a long run of earlier flights — the normalization of deviance. Columbia Magazine, <https://magazine.columbia.edu/article/challenger-disaster-normalization-deviance> ; University of Chicago Press blog, <https://pressblog.uchicago.edu/2016/01/07/the-normalization-of-deviance.html>
11. Ray DeGiorgio approved a GM ignition switch below torque specification and called it "the switch from hell" in an internal memo; a 2005 fix costing under a dollar a unit was rejected for

lacking an “acceptable business case” and logged as a customer-satisfaction issue rather than a safety one. Valukas Report, as reported by NBC News and House Energy & Commerce Committee documents, April 2014.

12. GM’s own board-commissioned internal investigation found no evidence of a coordinated conspiracy. CBC News, <https://www.cbc.ca/news/business/gm-ignition-switch-probe-finds-misjudgment-but-no-conspiracy-1.2664803> ; <https://www.cbc.ca/news/business/anton-valukas-tells-congress-there-s-no-coverup-at-gm-1.2679336>
13. A hundred and twenty-four deaths attributed to the GM ignition-switch defect. Feinberg-administered GM compensation fund, final figure as widely reported.
14. George Loewenstein’s hot-cold empathy gap: groundwork laid in a 1996 paper on visceral influences on behavior, the term itself formally named in a 2005 paper on medical decision making. Wikipedia, https://en.wikipedia.org/wiki/Hot-cold_empathy_gap ; PubMed, <https://pubmed.ncbi.nlm.nih.gov/16045419/> ; ResearchGate, https://www.researchgate.net/publication/7701109_Hot-Cold_Empathy_Gap_and_Medical_Decision_Making
15. Jon Elster on precommitment (Ulysses lashed to the mast); institutional review boards signing off before human-subjects research; the missile-launch two-officer/two-key rule; judicial warrants preceding searches. Jon Elster, *Ulysses and the Sirens* (Cambridge University Press, 1979); generically corroborated background regulatory practice, no single-source citation required per reviews/ch-10/verify.md item 12.
16. Peter Buxtun, U.S. Public Health Service venereal-disease investigator, learned around 1966 what the Tuskegee study actually was and filed a formal written protest in November 1966. Wikipedia, https://en.wikipedia.org/wiki/Peter_Buxtun ; Government Accountability Project, <https://whistleblower.org/>

uncategorized/whistleblower-peter-buxtun-and-the-tuskegee-syphilis-study/

17. Buxtun's second protest, filed in 1968, arguing that the men in the study didn't understand what was being withheld from them. Government Accountability Project, <https://whistleblower.org/uncategorized/whistleblower-peter-buxtun-and-the-tuskegee-syphilis-study/>
18. AP reporter Jean Heller broke the Tuskegee story in 1972, six years after Buxtun's first letter. Wikipedia, https://en.wikipedia.org/wiki/Peter_Buxtun ; Washington Post obituary, <https://www.washingtonpost.com/obituaries/2024/07/16/peter-buxtun-dead-tuskegee-syphilis/>
19. Class-action settlement (*Pollard v. United States*) reached in 1974 (corrected from an earlier draft's 1973, which was the year the suit was filed, not settled — see [reviews/ch-10/verify.md](https://www.fda.gov/oc/reviews/ch-10/verify.md) item 16). Justia, <https://law.justia.com/cases/federal/district-courts/FSupp/384/304/1370708/> ; LegalClarity, <https://legalclarity.org/the-tuskegee-syphilis-study-and-its-legal-consequences/>
20. The scandal directly produced the National Research Act of 1974 and the modern system of institutional review boards. The Hastings Center, <https://www.thehastingscenter.org/national-research-act-at-50-it-launched-ethics-oversight-but-it-needs-an-update/>
21. Cynthia Cooper, WorldCom vice president of internal audit, found roughly \$3.8 billion in operating expenses improperly capitalized, working largely at night. Tech Worker Handbook, <https://techworkerhandbook.org/media/cynthia-cooper-worldcom-2002/>
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23. Cooper named one of *Time's* Persons of the Year for 2002, alongside Sherron Watkins of Enron and Coleen Rowley of the FBI. *Time*, "The Whistleblowers," December 2002/January 2003 issue. <https://time.com/archive/6667965/persons-of-the-year-2002-the-whistleblowers/>
24. The 2036 prediction regarding a published bypass-reporting-channel policy among the twenty largest U.S. hospital systems by revenue; "twenty largest" is not a single stable ranking and should be checked against a named snapshot at adjudication. Becker's Hospital Review, <https://www.beckershospitalreview.com/rankings-and-ratings/40-largest-health-systems-in-the-us-2024/>; <https://www.beckershospitalreview.com/rankings-and-ratings/25-largest-health-systems-by-hospital-beds-2024/>
25. Robert Jackall, quoted: "What is right in the corporation is what the guy above you wants from you." Robert Jackall, *Moral Mazes* (Oxford University Press, 1988); wording and attribution confirmed via secondary sources, exact page citation unconfirmed. https://www.goodreads.com/book/show/279812.Moral_Mazes

Total: 25 endnotes.

Note: all three FAIL items from `reviews/ch-10/verify.md` (Boisjoly memo/"board of inquiry" misattribution, item 9; Loewenstein "named in a 1996 paper" overclaim, item 11; Tuskegee settlement year, item 16) were corrected in the frozen manuscript before this endnote pass; endnotes 3, 14, and 19 above cite sources for the corrected claims as they now read. Personal anecdotes (Priya, the author's own admission) are not drawn from the evidence packet and are not endnoted, consistent with the prologue's treatment of the Sarah material.

Sourced claims from `manuscript/frozen/prologue.md`, in order of appearance. Citations drawn from `reviews/prologue/verify.md`

(Gate 4 independent verification), not from the researcher's evidence packet.

1. Rachel Kudrna, identified as a fourth-year at Washington State University's Elson S. Floyd College of Medicine. AAMC News, "Applying to residency is stressful. Here's how experts hope to change that," September 13, 2022. <https://www.aamc.org/news/applying-residency-stressful-here-s-how-experts-hope-change>
2. Average U.S. medical graduate applies to around seventy residency programs, roughly double the figure a decade earlier. Carmody, J. B., Rosman, I. S., and Carlson, J. C. "Application Fever." *Cureus* 13, no. 3 (2021): e13804. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8034763>
3. Seventy applications costing about sixteen hundred dollars in ERAS fees, 2022–23 fee schedule (tiered: \$99 for applications 1–10, \$19 each for 11–20, \$23 each for 21–30, \$27 each for 31+, pricing seventy applications at \$1,599). Reconstructed 2022–23 ERAS tier schedule via Match A Resident, "Fees for the 2023-24 US Medical Residency Application Cycle." <https://blog.matcharesident.com/fees-for-the-2023-24-us-medical-residency-application-cycle/>
4. Interview travel for competitive specialties running past ten thousand dollars. Carmody, Rosman, and Carlson, "Application Fever," *Cureus* 13, no. 3 (2021): e13804. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8034763>
5. One program's estimated cost of \$3,736 in lost faculty clinical productivity per applicant interviewed. Carmody, Rosman, and Carlson, "Application Fever," *Cureus* 13, no. 3 (2021): e13804. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8034763>
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7. Rachel Kudrna, quoted: "There's tons of anxiety. It feels like after working so hard to become a physician, it could all still fall

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 10. AAMC (Association of American Medical Colleges) owns ERAS and charges a per-application fee. AAMC, “Fees for ERAS Residency Applications.” <https://students-residents.aamc.org/applying-residencies-eras/fees-eras-residency-applications>
 11. Program Signaling reduced applications per applicant in some specialties; dermatology applications falling from around seventy to about forty within two years (“by one tally” — figure genuinely contested across sources; the underlying direction and rough magnitude of a 2024–25 cycle-wide decline is well supported, but this specific dermatology pairing could not be pinned to one authoritative AAMC table). MedSchoolCoach, AAMC-derived Program Signaling application data by specialty, dermatology (no stable URL retained in the verification session; see `reviews/prologue/verify.md`, item 12, for the discrepancy with a second table showing dermatology at 72 applications in 2019, 57 in 2024, and 34 in 2026).
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14. Otolaryngology research items listed by successful (matched) applicants roughly doubling over six years, from about 8.4 to about 17.2. National Resident Matching Program, *Charting Outcomes in the Match: Senior Students of U.S. MD Medical Schools*, otolaryngology specialty data, 2016 and 2022 editions, as compiled during Gate 4 verification (no single stable URL retained; see [reviews/prologue/verify.md](https://www.nrmp.org/reviews/prologue/verify.md), items 14–15, where this figure replaced an unverifiable matched-vs.-unmatched claim originating in a mis-extraction from the evidence packet).
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17. Medical school enrollment continued expanding for roughly twenty years after the 1997 freeze (31% enrollment growth since 2002; 29 new accredited MD schools and 17 new DO schools since 2002). AAMC, enrollment press materials, 2017. <https://news.aamc.org/press-releases/article/enrollment-05252017/>

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Total: 18 endnotes.

Note: item 15 in `reviews/prologue/verify.md` (an unverifiable claim that the average unmatched applicant now lists more research activity than the average matched applicant did a decade earlier) was cut from the manuscript per the Gate 4 fix logged in `reviews/prologue/decisions.md` and does not appear in the frozen prologue; it is not endnoted here. Item 14 in `verify.md` (the "nobody in Washington unfroze" claim) was corrected in the frozen text to the 2021 Consolidated Appropriations Act thaw and is endnoted above as item 18.

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Appendix — Positions, Confidence, and What Would Change My Mind

Every substantive position the introduction volume takes: what it claims, how confident it is, and the strongest live objection against it. **Nothing in the book may state a position at a confidence higher than its rating here** (CANON.md §9, item 9; PRODUCTION_BIBLE.md §2.2).

Confidence scale — AXIOM / STRONG / MODERATE / PROVISIONAL. Ratings are deliberately conservative: where the honest call was ambiguous, this file rated lower rather than higher. “What would change my mind” is given for STRONG and MODERATE positions only, per assignment; AXIOM positions are chosen rather than argued for, and PROVISIONAL positions already carry an open alternative in their vulnerability entry.

Old-registry positions this file drops, downgrades, or otherwise revises relative to Positions/POSITION_REGISTRY.md are noted inline and summarized at the bottom.

I. Metaethics

POS-01: Phenomenal Value Realism

Confidence: MODERATE **CANON ref:** §2.1, §2.3 (Tier 1)

The badness of suffering is not a further fact appended to the experience; it is constitutive of it. What it is to be in pain is to be in a state whose felt character is bad. Moral epistemology is therefore partly a matter of introspective acquaintance with that felt badness rather than inference from evolved evaluative attitudes. This is Tier 1 of the degradation structure (POS-02) — the framework’s strongest metaphysical claim about value, made without the three-independent-layers argument the prior draft used and CANON now bans.

Known vulnerability: Heathwood and Feldman’s attitudinal theories hold that badness lies in the subject’s dislike of a sensation, not in the sensation’s felt character itself. Pain asymbolia — patients who report the sensation without its badness — is compatible with either reading: the attitudinal theorist can say the missing element is the dislike, not a constitutive property. CANON concedes this reading is “live” (§2.4) and commits only to showing the framework survives either way — a demonstration that does not appear in CANON itself and is presumably left to Ch. 5. (Heathwood, Feldman, per CANON §2.4.)

What would change my mind: A worked demonstration — currently absent from CANON — that the framework’s downstream architecture (rules, override, aggregation) survives unchanged under the attitudinal reading; or evidence that pain asymbolia is better explained by loss of attitude than by loss of phenomenal character.

POS-02: The Degradation Tiers (Practical Invariance)

Confidence: STRONG CANON ref: §2.3

A three-tier structure replaces the banned three-independent-layers defence. Tier 1: phenomenal realism true — full strength. Tier 2: illusionism true — representationalism about valence, weaker metaphysics, identical normative output. Tier 3: strong anti-realism true — the framework degrades into quasi-realism plus a coordination theory; “constrained by objective reality” is lost, but every institutional and practical conclusion survives. The claim is that the book’s *practical content* is invariant across the live space of metaethical positions, not that any one metaethical position is proven.

Known vulnerability: Tier 3 is the load-bearing claim and the least demonstrated one. Quasi-realism (Blackburn) earns realist-sounding talk within a projectivist semantics, but whether quasi-realist “oughts” can bear the weight the override architecture and rights structure place on them — action-guiding force for a reader who has already accepted anti-realism — is a live dispute inside metaethics itself (“creeping minimalism”; Dreier vs. Blackburn). CANON asserts the invariance; it does not yet show it worked through a case.

What would change my mind: A demonstration, chapter by chapter, that a convinced error theorist reading only Tier-3 material would still find the override conditions and rights architecture action-guiding rather than merely descriptive of an equilibrium — or a documented instance of a Tier-3 reader failing to find the material binding.

II. The Axiom and Its Scope

POS-03: The Axiom

Confidence: AXIOM CANON ref: §3

The suffering of a conscious being matters. Chosen, not derived — this is the only place normativity enters the system. The scope is “a conscious being,” not a person, a citizen, or a member of any in-group; this scope claim is what makes the circle-expansion argument (POS-05) non-arbitrary once the axiom is granted.

Known vulnerability: As an axiom it isn’t open to the usual “prove it” objection, but the honest cost of any chosen starting point is that a rival axiom — a perfectionist axiom, a preference-satisfaction axiom, a Nietzschean value-creation axiom — can be chosen with equal formal legitimacy. CANON offers no argument that this axiom is uniquely rationally compelled over rivals, only that it is the one the book runs on. This is the axiom-level version of the same worry Sidgwick raises one level up, at impartiality (POS-04).

POS-04: The Coordination Derivation of Impartiality

Confidence: MODERATE CANON ref: §3

The derivation of impartiality splits from the axiom rather than being derived from it directly (the prior draft’s burden-shifting move is banned, CANON §9 item 5). The axiom supplies the currency: valence is what has value. The coordination layer supplies the impartiality: agents in an indefinitely iterated game, uncertain of their future bargaining position, converge on norms that count valence impartially, because partial norms are not stable equilibria among agents who cannot guarantee staying on the

favoured side. CANON states this version avoids Gauthier's fatal problem because the base is value rather than preference satisfaction.

Known vulnerability: Sidgwick's dualism of practical reason — the indexical egoist's reply that CANON explicitly says the book "must not pretend" doesn't exist (§3). "Uncertain of future bargaining position" is doing the entire derivational work, and it is an empirical premise about the agent's beliefs, not an entailment of the axiom: an agent confident of remaining durably advantaged (an entrenched elite, a hegemon) has no obvious reason, on this argument alone, to converge on impartial norms rather than defect from them. This is Gauthier's own unresolved "why would the powerful bargain" problem, restated rather than dissolved by moving from preferences to value.

What would change my mind: A demonstration that even durably-advantaged agents have principled reason for uncertainty about their future position — e.g., an argument from the instability of all positions of advantage over sufficiently long horizons. Without that, the derivation answers the egoist who is genuinely uncertain, not the egoist who reasonably believes they aren't.

POS-05: The Unstable-Schelling-Point Circle Argument

Confidence: STRONG **CANON ref:** §3

For beings who cannot reciprocate — infants, animals, the severely disabled, future people — the framework argues that arbitrary moral-circle boundaries are unstable Schelling points. Every boundary short of "all valence" needs a defensible line; every line ever proposed (tribe, race, nation, sex, species, cognitive

threshold) has eroded under pressure because none is a natural focal point and each is exploitable by anyone with an interest in redrawing it. “All valence” is the only boundary that doesn’t erode. Circle expansion is equilibrium selection, not moral fashion. CANON calls this the framework’s most original argument.

Known vulnerability: The equilibrium-selection logic is strongest for boundaries drawn among agents who can plausibly threaten, bargain, or shift coalitions — reciprocity-capable populations. Beings who can never reciprocate don’t exert the coalition pressure that makes a boundary “unstable” in the game-theoretic sense; the argument may explain why humans keep redrawing lines among humans without yet showing why a stable coalition of reciprocity-capable agents is rationally compelled to extend the boundary to those who can never join the bargain. This is the general difficulty contractarian ethics has always had with “marginal cases” (cf. the standard objection to Rawlsian and Gauthier-style contractarianism).

What would change my mind: A version of the argument that shows instability even for boundaries drawn against permanently non-reciprocal beings — e.g., that the demonstrated exploitability of a line in one case is evidence about the line-drawer’s reliability in general, independent of the excluded party’s bargaining power.

III. Value Theory

POS-06: Value Monism, Goods Pluralism

Confidence: MODERATE **CANON ref:** §4

Only valence is intrinsically valuable. Autonomy, competence, connection, understanding, and meaning are robustly necessary *conditions* of sustained positive valence in beings constituted like

us — empirically load-bearing, not axiomatically independent. This preserves the single-axiom claim and replaces the banned six-dimensional value vector (CANON §9 item 4).

Known vulnerability: Nozick’s experience machine is the classical challenge to any valence-monist theory: if only valence mattered, a life of maximal felt wellbeing produced by deception should be equally or more choiceworthy than a real one with equal or lesser valence. Most people’s considered judgment says otherwise, which suggests autonomy, connection, and understanding are valued for more than their contribution to valence. Objective-list theorists (Parfit, Finnis, and Nussbaum in her own voice rather than as a floor-mechanism) hold these goods matter independently.

What would change my mind: A worked answer to the experience machine that doesn’t quietly reintroduce an independent value for authenticity or reality-contact (which would violate the single-axiom claim) — or evidence that the anti-machine intuition is itself best explained by an unreliable status-quo bias rather than a real, independent value.

POS-07: Prioritarianism

Confidence: STRONG **CANON ref:** §4

The moral weight of a valence increment rises as the recipient’s baseline falls. Blocks the utility monster: the monster’s marginal gains are weighted down (Parfit, “Equality or Priority?”; Arneson).

Known vulnerability: The weighting function — how fast priority rises as baseline falls — is not derived from anything else in the framework and is chosen for the verdicts it produces. A sufficiently adversarial monster case (baseline low enough, or

marginal-gain function steep enough) can in principle be constructed to defeat any finite weighting. Separately, prioritarianism is criticized for smuggling in an aggregative, impersonal calculus that the separateness-of-persons material (POS-09) says shouldn't be doing this work (Otsuka & Voorhoeve).

What would change my mind: A principled, non-stipulated derivation of the weighting function from the coordination-theoretic foundations, rather than a function chosen to produce the desired verdicts; or a demonstration that no finite weighting can be defeated by an adversarially constructed monster case.

POS-08: The Sufficientarian Threshold

Confidence: MODERATE **CANON ref:** §4

A capability floor, lexically prior to aggregate maximisation below the line (Crisp; Nussbaum; Frankfurt).

Known vulnerability: The “why this threshold and not a slightly higher or lower one” arbitrariness objection (Casal, “Why Sufficiency Is Not Enough”) and the discontinuity problem: lexical priority means a person a hair's-breadth below the line gets absolute priority over vastly larger gains to people just above it, which sits awkwardly against the framework's own claim that valence is a single fungible currency (POS-06) — a marginal unit of valence shouldn't flip from negligible to lexically dominant based on which side of a line it falls on.

What would change my mind: A principled, non-stipulated derivation of where the threshold sits, rather than an intuitively chosen number; or a demonstration that the discontinuity is not action-guiding in any case the book actually relies on.

POS-09: Separateness-of-Persons Constraints

Confidence: PROVISIONAL CANON ref: §4

The transplant case is blocked by a rights structure derived from coordination stability, not by welfare arithmetic.

Known vulnerability: Instrumentally-grounded rights are, in principle, revisable whenever the coordination calculus that generated them changes — which is exactly what rights are supposed to prevent. A right that holds “because it’s currently the stable equilibrium” is not obviously a right at all on the classical understanding of rights as trumps against precisely this kind of recalculation (Dworkin, *Taking Rights Seriously*). This is the same unresolved objection the old registry logged against its own rights position (POS-P07, PROVISIONAL/HIGH) — CANON’s revision does not supply new grounding beyond restating the coordination-stability claim, and no chapter in the current 18-chapter brief is dedicated to developing it further (see CANON-KEEPER QUESTION 4, below).

POS-10: The Person-Affecting Default on Population Ethics

Confidence: PROVISIONAL CANON ref: §4, §8

The book takes a position on the repugnant conclusion rather than deferring: the default is a person-affecting restriction, with its costs acknowledged. Alternatives (critical-level views, or biting the bullet per Huemer/Tännsjö) are acceptable if the Showrunner prefers, but silence is not. CANON itself lists population ethics among what the framework does not solve (§8): “every welfarist theory bleeds here.”

Known vulnerability: Person-affecting views standardly generate the non-identity problem and, per Parfit’s mere addition paradox, can produce intransitive rankings across cases that differ only in whether the same or different people exist (Parfit, *Reasons and Persons*, Part IV) — the same kind of instability a book built on “unstable Schelling points” (POS-05) should be uncomfortable defending elsewhere. CANON’s own hedge (“alternatives are acceptable if the Showrunner prefers”) signals this is not yet a settled position — see CANON-KEEPER QUESTION 3.

IV. The Diagnosis

POS-11: The Three-Axis Diagnosis

Confidence: STRONG **CANON ref:** §5

The prior draft’s single diagnosis (coordination failure / “Moloch”) is replaced with three axes, each with a matched tool: Axis 1, coordination failure (mechanism design); Axis 2, circle failure — successful coordination *against* excluded parties, where better incentive alignment among perpetrators makes things worse (rights, constitutional constraint, circle-expansion arguments); Axis 3, epistemic failure (the epistemic commons, free expression, forecasting institutions). CANON presents this as a genuine typological contribution.

Known vulnerability: The partition may not be clean in practice. Many real cases are compound — a cartel is simultaneously a coordination success among members, a circle failure against outsiders, and sustained by epistemic failure among the excluded public. A taxonomy that requires assigning a dominant axis to compound cases risks inconsistent application by different readers using the same method, which cuts against the

method-not-policy commitment's (POS-22) promise of faithful convergence.

What would change my mind: If the book's worked examples show two careful readers routinely assigning the same real case to different axes and reaching different tool recommendations, the taxonomy needs explicit compound-case handling, or needs to be presented as overlapping rather than as a clean partition.

V. The Decision Architecture

POS-12: The Rules-First Default (internal label "Mode A" — see Glossary)

Confidence: STRONG **CANON ref:** §6.1

Follow pre-committed rules and cultivated virtues by default. Justification: rules are bias-resistant, cognitively cheap, predictability-producing (which is what makes an agent trustworthy), and they encode base rates the inside view can't access — even ideal consequentialists would adopt rules, because a rule-follower can be trusted in ways a calculator cannot. Chesterton's fence: an inherited institution is evidence; revision requires understanding why the rule exists first.

Known vulnerability: Particularism (Dancy) denies that general rules can track the morally relevant features of situations at all — on this view, good judgment is irreducibly case-by-case, and the "trust" argument for rules is a fact about how rule-following *looks* to others, not a fact about what's actually right to do, treating two different questions as one. CANON does not currently name Dancy or particularism as an interlocutor anywhere in the read material.

What would change my mind: A case, addressed in the book, where competent particularist judgment reliably outperforms rule-following even once the trust and predictability costs of case-by-case reasoning are priced in.

POS-13: The Five Override Conditions

Confidence: STRONG CANON ref: §6.2

Catastrophic irreversible harm from rule-following; harm attributable to the rule rather than merely coincident; last resort with alternatives genuinely exhausted; articulable to a skeptical third party who does not share your interests; self-interest absent or explicitly corrected for.

Known vulnerability: The “articulable to a skeptical third party” test is self-administered — applied by the agent’s own imagination of that third party, in the moment. Nearly everyone who has ever wrongly overridden a rule believed their case would survive this test; Milgram’s participants largely believed their *compliance*, not their resistance, was the defensible position, which cuts against confidence-in-articulability as a filter in either direction. The three procedural gates (POS-14) address the *timing* of the test but not this self-administration problem.

What would change my mind: Evidence that pairing the conditions with the procedural gates (mandatory deliberation, genuine rather than imagined third-party consultation) measurably reduces the false-positive override rate compared to the conditions applied alone.

POS-14: The Three Procedural Gates

Confidence: STRONG CANON ref: §6.2–6.3

Applied *before* the five conditions are consulted: a bright-line trigger check (imminent large-scale loss of life / irreversible systemic damage / institutional capture disabling normal error-correction); a mandatory deliberation period; and mandatory documented post-hoc review regardless of outcome. Sealed or delayed documentation, with an exemption from the delay for time-sensitive institutional capture, resolves the whistleblower contradiction (§6.3). CANON calls the procedural gating the framework’s single most original contribution, and the chapter brief for Ch. 10 independently confirms it as the book’s best-written, most original chapter.

Known vulnerability: The bright-line trigger still requires judgment — “institutional capture that disables normal error-correction” is not self-identifying, and the agent applying the gate is the same fallible evaluator the gates exist to constrain. This risks a regress one level up rather than an elimination of it: instead of “is my override justified,” the agent now asks “does my situation trigger the gate,” and the two questions are not obviously easier to answer honestly under motivated reasoning.

What would change my mind: A demonstration, via the historical cases the book uses, that gate-triggering judgments are in fact more legible to outside observers than override judgments were — this is the empirical claim the architecture depends on, and CANON asserts it without yet showing it worked through a case.

POS-15: The False-Negative / Role-Relativity Doctrine

Confidence: PROVISIONAL CANON ref: §6.3

Every condition and gate above filters false positives (overriding when you shouldn’t). The framework must also

address false negatives — complying when you shouldn't. Milgram, My Lai, and institutional cover-ups are false-negative failures. Additions: expected-cost calibration by domain (private/reversible stakes favour heavy anti-override bias; captured-institution/irreversible stakes invert it); role-relative thresholds (a physician, a soldier, a civil servant, and a private citizen have different override profiles because their rules have different justificatory bases and capture risks); and the sealed/delayed-documentation fix for time-sensitive capture. CANON itself flags this material “NEW, must be in the book” — freshly added, not yet load-tested the way the override conditions are.

Known vulnerability: Role-relative thresholds risk being exactly the special-pleading structure the false-negative material exists to prevent from the other direction — “my role justifies a lower threshold” is structurally identical to “my case is special,” the self-serving reasoning pattern the whole architecture filters against. CANON does not yet specify who sets the role-relative thresholds or how they resist being set by the role-holders themselves. This is an internal tension rather than a literature citation, flagged honestly because CANON supplies no rebuttal in the material available.

POS-16: Virtues as Firmware, Humility as Keystone

Confidence: MODERATE CANON ref: §6.4

Virtues are installed response patterns the architecture runs on: epistemic (curiosity, relinquishment, lightness, evenness, steel-manning, empiricism, simplicity, humility) and instrumental (effortful improvement, precision, scholarship, the void). Humility is the keystone; without it every other virtue becomes a weapon.

Known vulnerability: Situationism (Doris, *Lack of Character*) presents substantial experimental evidence that behavior is driven more by situational variables than by stable cross-situational character traits, which threatens the coherence of “installed response patterns” as psychological reality rather than aspiration. The Ch. 11 brief itself flags that this objection needs “the honest reply,” indicating it is not yet answered in the material available.

What would change my mind: The honest reply the chapter brief calls for, actually worked through — e.g., evidence that virtues-as-trained-defaults (deliberately built habits with situational triggers) survive the situationist critique better than virtues-as-stable-traits do, which is a different and more defensible claim than the one situationism actually targets.

VI. Applications

POS-17: Error-Correction Primacy

Confidence: STRONG CANON ref: §7

Design institutions for the ability to detect and reverse mistakes, not for efficiency or for being right the first time. Separation of powers, federalism, subsidiarity, antifragility. Democracy (POS-19) is justified on this ground.

Known vulnerability: Some errors are irreversible by the time they’re detected (existential and near-existential risk), and error-correction as the primary design criterion has little to say to a threat that must be gotten right the first time because there is no “after” in which to correct it. The old registry’s rebuttal to this exact objection (POS-P01) punted to a dedicated existential-risk chapter; that chapter is not part of this introduction volume’s scope — it is quarantined to Volume II under the method-not-

policy commitment (POS-22) — so the rebuttal currently has no home in this book.

What would change my mind: An explicit acknowledgment in the Coda that error-correction primacy has a scope limit at irreversible catastrophic risk (consistent with §8’s honesty mandate) — or a demonstration that error-correction-designed institutions do better even on first-strike-only threats, e.g. by reducing the odds of ever facing one.

POS-18: Subsidiarity

Confidence: MODERATE **CANON ref:** §7

Decisions should be made at the lowest level of governance competent to handle them.

Known vulnerability: Problems with genuine cross-jurisdictional externalities (climate, pandemics, frontier AI) escalate upward on CANON’s own account, but “genuine externality” is precisely the contested judgment call in most real disputes. The escalation rule can be invoked by centralizers to justify almost any upward transfer of authority and denied by localists to resist almost any oversight; CANON supplies no test for adjudicating contested cases, only the uncontested-in-principle rule. This mirrors the old registry’s own unresolved objection (POS-P05).

What would change my mind: A worked example — ideally the Ch. 17 divergence demonstration itself — showing the externality test actually discriminating between a genuine escalation case and an invoked-but-spurious one.

POS-19: Democracy as Removal Mechanism, Not Oracle

Confidence: STRONG CANON ref: §7

Democracy is justified as an error-correction mechanism, not as an oracle — it is good at removing bad rulers and bad at technical questions.

Known vulnerability: Removal-mechanism competence and minority protection are different achievements. A democracy can be excellent at removing bad rulers while a stable majority uses that same removal power against a minority repeatedly and lawfully — the tyranny-of-the-majority problem (de Tocqueville, Mill; old registry POS-P04). “Good at removing bad rulers” doesn’t by itself explain why the removed rulers’ policies toward a minority were wrong, since majority preference doesn’t track that. CANON does not show this justification connected to the separateness-of-persons/rights material (POS-09) that would presumably do the minority-protecting work — see CANON-KEEPER QUESTION 5.

What would change my mind: An explicit account of how the removal-mechanism justification connects to the rights constraints elsewhere in the architecture, closing the gap the vulnerability above identifies.

POS-20: The Knowledge Problem and the Four Market Failure Classes

Confidence: STRONG CANON ref: §7

Hayek’s knowledge problem: prices carry dispersed information no planner can assemble. Markets fail on externalities, public goods, information asymmetry, and market power. Remedy

matched to failure type: Pigovian correction where measurable, Ostrom-style community governance where the commons is local and users can monitor each other, state provision for genuine public goods.

Known vulnerability: Pigovian correction requires accurate measurement of the externality, which is frequently unavailable or politically contestable, and the remedy is vulnerable to the same capture dynamics the framework's own Axis 2 (circle failure, POS-11) diagnoses elsewhere — a regulator empowered to set a corrective tax or subsidy is a target for exactly the rent-seeking the knowledge-problem argument warns markets and planners both are vulnerable to. Restates the old registry's own honestly-logged objection (POS-EC02).

What would change my mind: A worked case — the Ch. 17 divergence demonstration is the natural home — where measurement difficulty and capture risk are priced into the method itself rather than assumed away.

POS-21: The Capability Floor

Confidence: MODERATE **CANON ref:** §7

The Sen/Nussbaum capability floor serves as the redistributive commitment in the markets-and-institutions material.

Known vulnerability: Two objections, stacked. First, the standard objection to capability approaches: which capabilities count and how they're weighted is substantially stipulated (Nussbaum's list) rather than derived — an underspecification problem structurally similar to the sufficientarian threshold's arbitrariness (POS-08; Wolff & De-Shalit's critique of capability lists). Second, and specific to this framework: CANON asserts value monism (POS-06) — only valence is intrinsically valuable,

other goods are instrumental conditions of it — but states the capability floor as “the redistributive commitment” without showing it is derived from valence rather than smuggled in as a second, independent axiological commitment. This second point is not resolved here — it would require inventing an argument CANON doesn’t supply — and is flagged as CANON-KEEPER QUESTION 1.

What would change my mind: An explicit derivation, in CANON or in the Ch. 17 material, of the capability floor from the valence-monism foundation — showing which capabilities are floor-worthy because of their instrumental necessity for sustained positive valence, rather than the current unranked list.

POS-22: The Method-Not-Policy Commitment

Confidence: STRONG CANON ref: §7

Present the method, not the policy bundle, in both the institutions and markets chapters. Specific policy conclusions (minimum wage, school choice, zoning, carbon pricing) go to Volume II with their empirical premises exposed. The introduction must demonstrate that two readers with different defensible empirical priors can apply the method faithfully and diverge — and must show what evidence would settle it.

Known vulnerability: The method is not actually value-neutral between political traditions even where it’s empirically open. Prioritarianism (POS-07), the sufficientarian/capability floor (POS-08, POS-21), and error-correction-as-primary (POS-17) are all substantive normative commitments baked into the method itself, not merely empirical premises left open for the reader to fill in. A libertarian reader who rejects redistributive priority weighting isn’t disagreeing about facts — they’re disagreeing with the

method's normative content, which the "faithful divergence" framing may undersell.

What would change my mind: The Ch. 17 divergence demonstration actually producing readers who span a wide enough range of political priors — not just disagreement within a shared redistributive consensus — to show the normative commitments aren't doing more work than the "method" framing admits.

What was dropped or downgraded from the old registry, and why

- **POS-M01 (Constructivist Realism)** — dropped entirely. The name and the framing are banned (CANON §9, items 1–2); replaced by POS-01/POS-02.
- **POS-M02 (Valence Realism, STRONG/LOW via the three-layer defence)** — the three-layer grounding is banned (CANON §9 item 2; CANON §2.2 explains why it fails: one objection reaches every layer). Its replacement, POS-01, is rated MODERATE rather than carrying forward the old STRONG/LOW rating, because the degradation-tier strategy is honest about *not* claiming the old distributed-burden robustness.
- **POS-A01 (Dual-Process Decision Making, downgraded to LOW vulnerability after the gating protocol was added)** — carried forward as POS-12/13/14 but *not* carrying forward the LOW-vulnerability optimism. This file keeps STRONG confidence (the architecture is well-argued) but restates the self-administration and regress problems as live rather than resolved, since CANON's revision doesn't add material that closes them.

- **POS-A02 (Morality as Coordination Technology, Bounded)** — superseded by the three-axis diagnosis (POS-11), which is a genuine typological improvement over the old two-axis-plus-three-exceptions structure.
 - **POS-P07 (Rights as Coordination Equilibria, PROVISIONAL/HIGH, flagged “NEEDS SIGNIFICANT DEVELOPMENT”)** — carried forward essentially unchanged as POS-09, still PROVISIONAL, because CANON’s revision does not supply the development the old registry called for and no chapter in the current brief set is dedicated to it (CANON-KEEPER QUESTION 4).
 - **POS-P03 (Strong Free Speech Default) and POS-EC04 (Prediction Markets)** — not carried forward as standalone positions. They’re absorbed into POS-11’s Axis 3 (epistemic failure) tool set rather than argued independently anywhere in the read material; if the book develops them further as free-standing claims, they should be split back out.
 - **POS-F01–F03 (existential risk, cautious AI, consciousness-based moral circle)** — not carried forward. These belong to chapters (old CH-31, CH-32, CH-34) that are outside the 18-chapter introduction-volume scope confirmed in `spec/CHAPTER_BRIEFS.md`. The Coda (§8) touches consciousness and blindness in general terms, which is covered implicitly by POS-01/02’s scope rather than as a separate position.
 - **POS-EC05 (Fiscal Conservatism with Social Investment)** and other single-policy positions (minimum wage, etc.) — dropped, per CANON §9 item 7 (“specific policy conclusions presented as derivations” must not appear) and the method-not-policy commitment (POS-22).
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CANON-KEEPER QUESTIONS

1. **Capability floor derivation.** CANON §4 commits to value monism (only valence is intrinsically valuable) and CANON §7 states the capability floor “as the redistributive commitment” without connecting the two. Is the capability floor meant to be derived instrumentally from valence (per POS-06’s logic), or is it intended as an independent distributive commitment that sits outside the single-axiom claim? POS-21’s confidence rating depends on this answer.
2. **The attitudinal-theory demonstration.** CANON §2.4 commits the book to “concede it’s live; show the framework survives on either reading” for the Heathwood/Feldman objection, but no such demonstration appears in CANON itself. Is this demonstration planned for Ch. 5’s drafting, or does Canon Keeper want to pre-approve its content before drafting begins, given it isn’t yet canonical anywhere? POS-01’s confidence could rise from MODERATE to STRONG if this is resolved in text.
3. **Population ethics — is the person-affecting default locked?** CANON §4 explicitly leaves the repugnant-conclusion answer open to Showrunner preference (“alternatives... are acceptable if the Showrunner prefers”). POS-10 defaults to the person-affecting restriction for rating purposes because that’s CANON’s stated default, but if the Showrunner has since chosen an alternative (critical-level view, or biting the bullet), this file needs updating before Ch. 10/Coda drafting locks in the choice.
4. **No dedicated rights chapter.** The old TOC had a chapter developing “Rights as Coordination Equilibria” (old CH-19); the current 18-chapter brief set (spec/CHAPTER_BRIEFS.md) has no equivalent — the closest

material is split thinly across Ch. 6 (“Whose Suffering”) and Ch. 16 (“Institutions”). POS-09’s separateness-of-persons claim and the transplant-case rebuttal in CANON §4 are asserted without a chapter that develops them. Is this intentional (the rights material is meant to stay compressed), or is a chapter or chapter-section missing from the brief set?

5. **Democracy and minority protection.** CANON §7 justifies democracy purely as a removal mechanism and doesn’t connect that justification to the rights/separateness-of-persons constraints (§4) that would presumably protect minorities from majority removal-power abuse. Should Ch. 16 make this connection explicit, or is it deliberately left for the reader to infer from Book-wide context?
6. **Mode A’s English replacement.** CANON §6.1 and the Ch. 8 brief both instruct that “Mode A” and “Mode B” must not appear in the book text (“find English for them”). CANON §6.2 already supplies a partial answer for Mode B — it is headed “Mode B — the override,” so “the override” is CANON’s own working synonym. No equivalent synonym is given anywhere in the read material for Mode A. Should GLOSSARY.md propose a placeholder term (e.g., “the default,” “the standing rule”) for cross-team use before Ch. 8 is drafted, or should Canon Keeper leave this entirely to the Drafter?